

SIP AND IMS · STUDY GUIDE

SIP and IMS, explained

A visual guide for telecom engineers: SIP messages, call flows, SDP, RTP, IMS and VoLTE.

Telecom Knowledge Hub

Nishant Tyagi · nishanttyagi.de

Contents

01	SIP Basics	8 steps
02	SIP Message Structure	8 steps
03	Responses, Transactions and Dialogs	8 steps
04	The Basic Call Flow	8 steps
05	SDP and Offer/Answer	8 steps
06	RTP and RTCP	8 steps
07	Registration and Authentication	8 steps
08	IMS Architecture	8 steps
09	IMS Registration	8 steps
10	The VoLTE Call Flow	8 steps
11	IMS Services	8 steps
12	Transport, Security, NAT and Troubleshooting	8 steps

SIP Basics

What SIP is, what it is not, and the elements of a SIP network.

1.1 What SIP does

SIP does

- Locate the user
- Invite to a session
- Modify and end the session
- Register where a user is

SIP does not

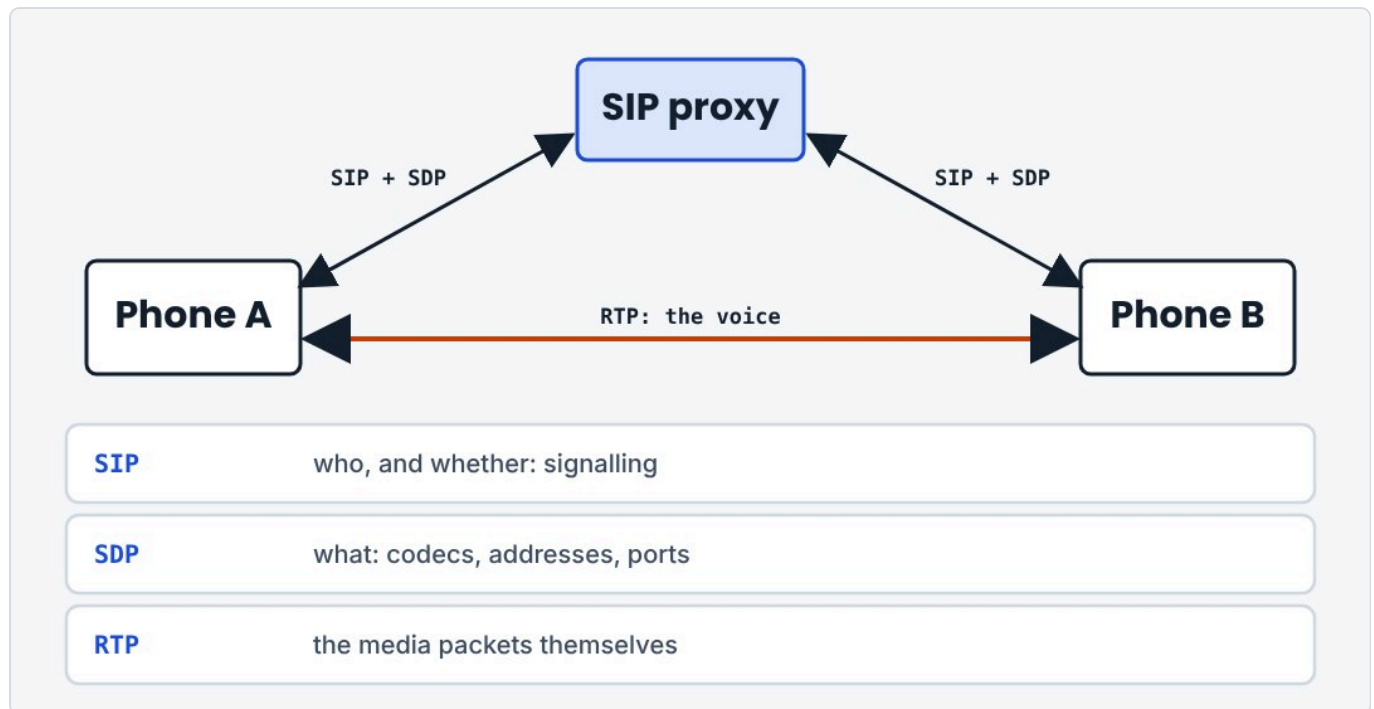
- Carry voice or video (RTP does)
- Describe the media (SDP does)
- Guarantee quality (the network does)
- Define the services themselves

A text protocol, modelled on HTTP and e-mail

SIP, the Session Initiation Protocol, sets up, changes and ends sessions between users. A session is usually a voice or video call. SIP finds the other party, agrees that a session should exist and tears it down. It does not carry the voice itself and it does not describe the media. Other protocols do that.

RFC 3261 §1, §2

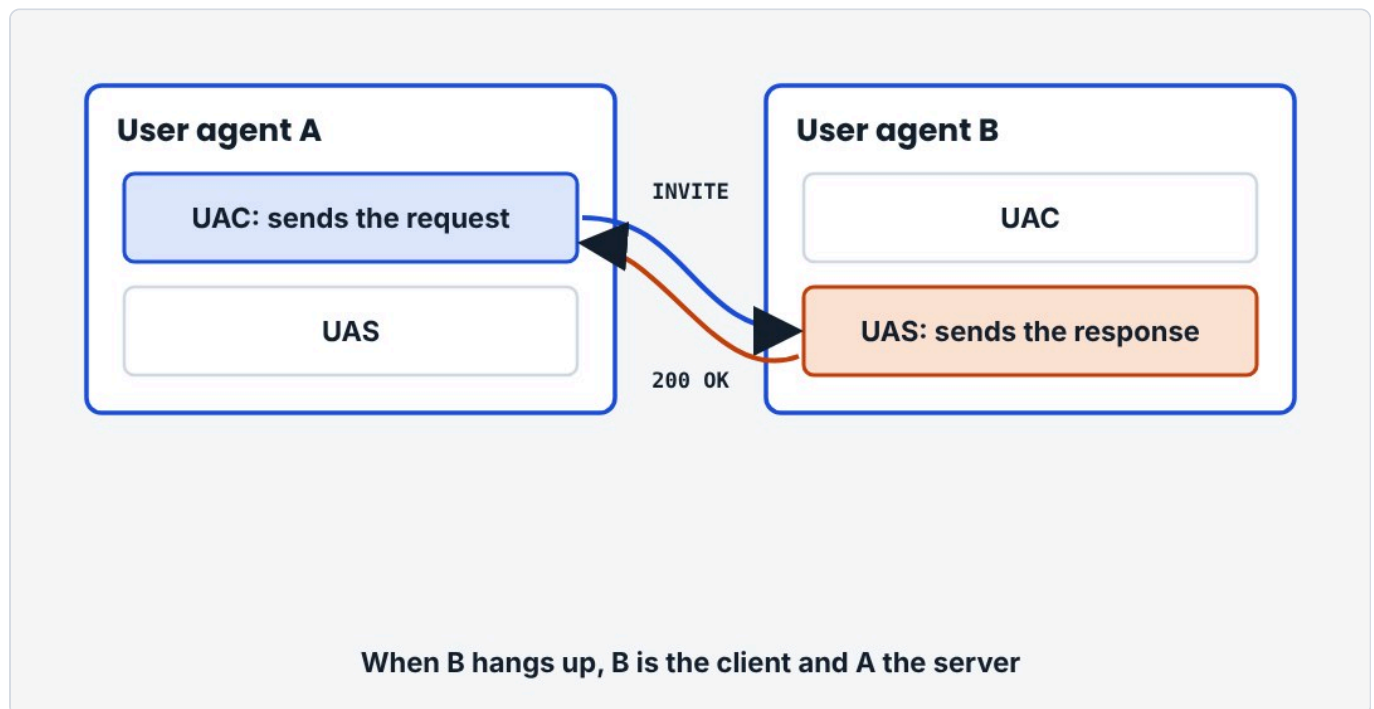
1.2 Three protocols work together



A call needs three protocols. SIP carries the signalling. Inside the SIP message body, SDP describes the media: codecs, IP addresses and ports. Once both sides agree, the voice flows as RTP packets, usually directly between the endpoints and not along the signalling path.

RFC 3261 · RFC 8866 (SDP) · RFC 3550 (RTP)

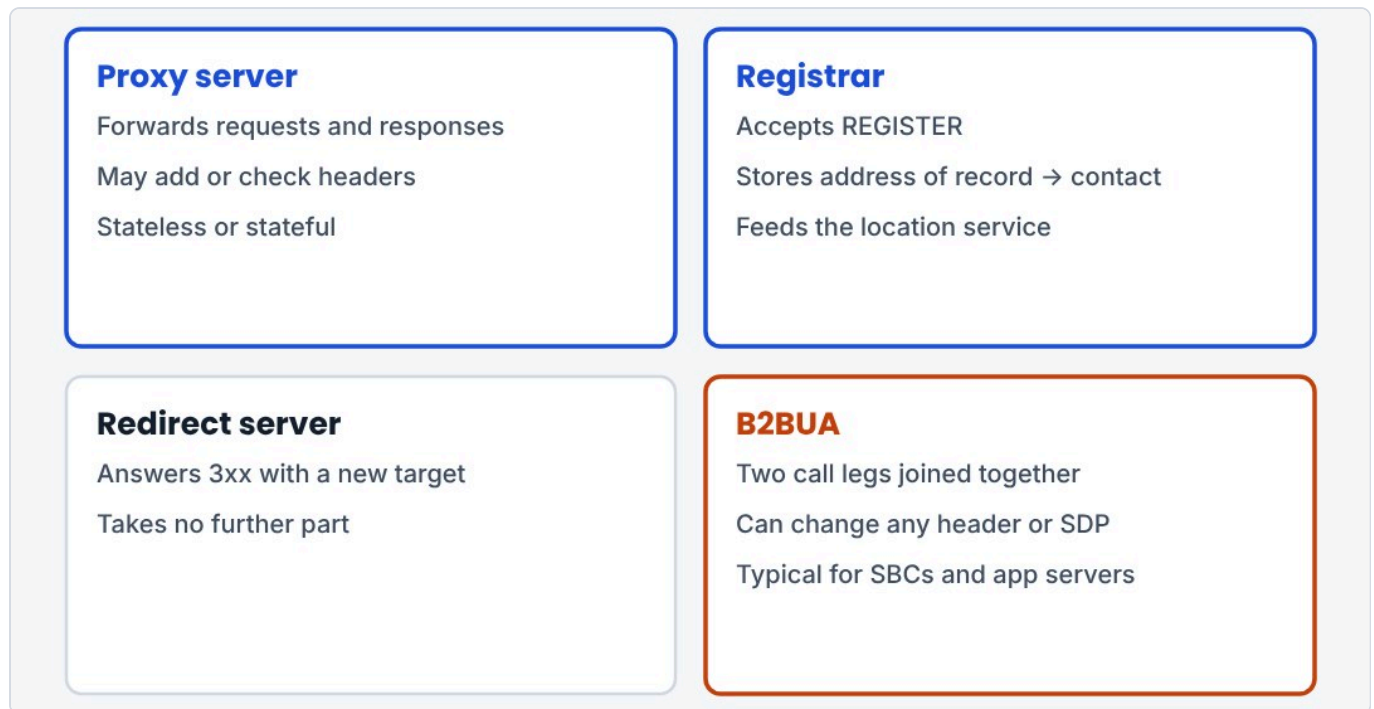
1.3 User agents



The endpoints are user agents. Each has two halves. The user agent client sends requests and the user agent server answers them. A phone is a client when it calls and a server when it is called. The roles last for one transaction only.

RFC 3261 §6, §8

1.4 Servers in the network



Between the user agents sit servers. A proxy forwards requests towards the destination. A registrar stores where each user can currently be reached. A redirect server answers with a new address instead of forwarding. A back-to-back user agent ends one call leg and starts another, which gives it full control.

RFC 3261 §6, §16, §10, §8.3

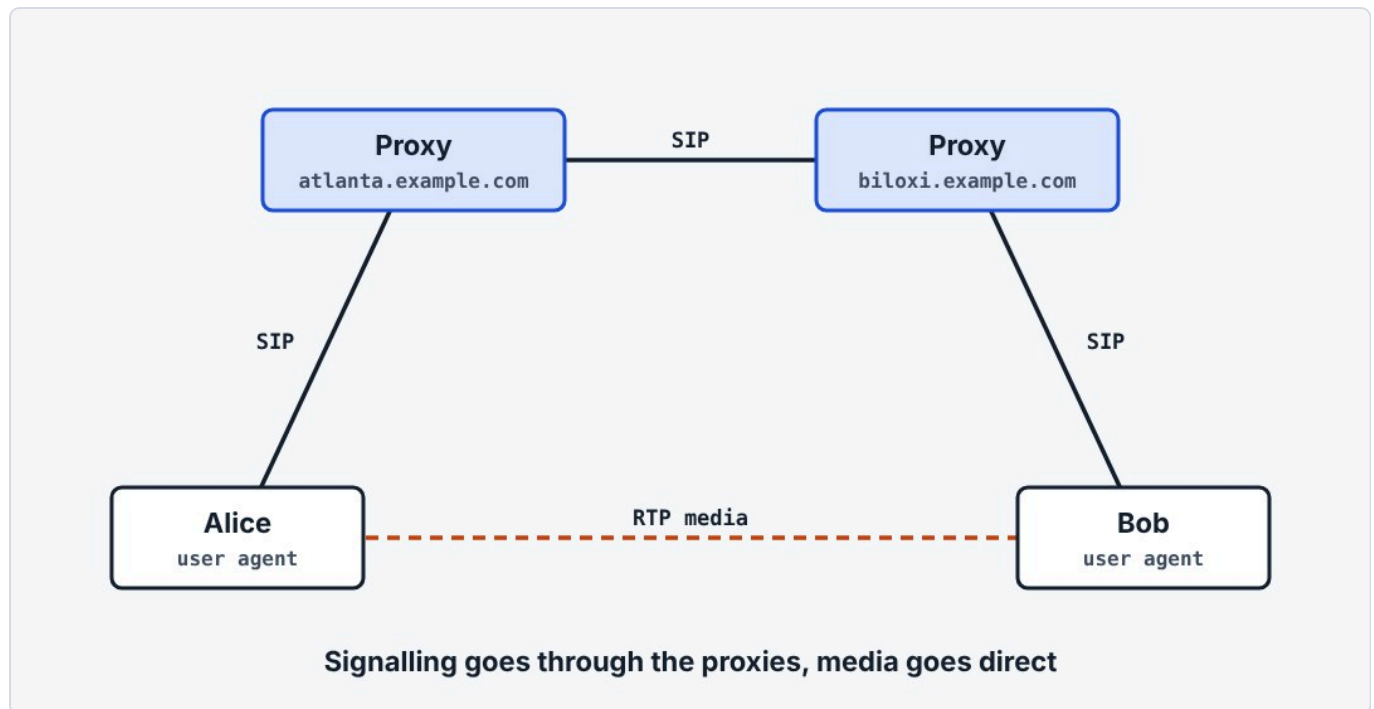
1.5 SIP addresses

FORM	EXAMPLE
SIP URI	<code>sip:alice@example.com</code>
Secure SIP URI	<code>sips:alice@example.com</code>
With port and transport	<code>sip:alice@192.0.2.10:5060;transport=tcp</code>
Number as SIP URI	<code>sip:+4930123456@ims.example.com;user=phone</code>
tel URI	<code>tel:+4930123456</code>
Address of record	the public identity, for example sip:alice@example.com
Contact	where the device is now, for example sip:alice@192.0.2.10

A user is named by a SIP URI, which looks like an e-mail address. The address of record is the public, stable name. The contact is the device where the user is right now. Registration binds the two. Telephone numbers are written as tel URIs or as SIP URIs with a user=phone parameter.

RFC 3261 §19.1 · RFC 3966

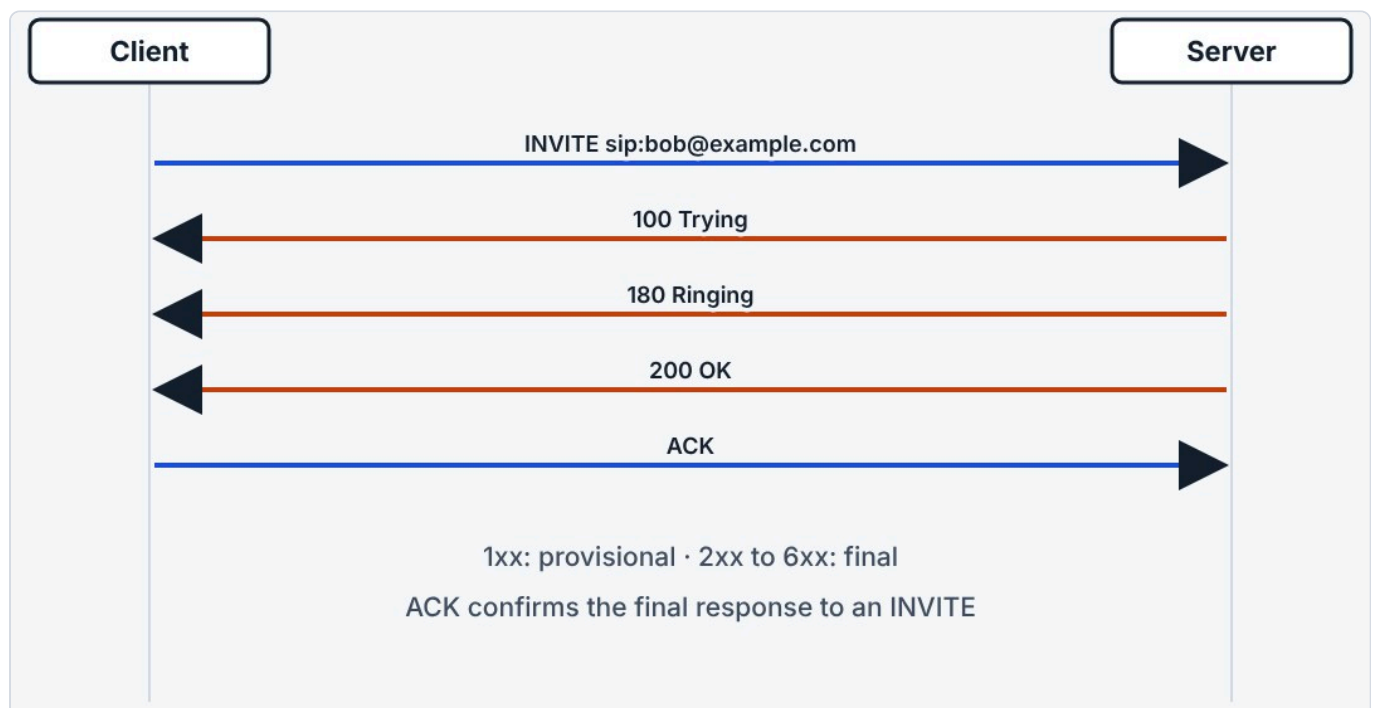
1.6 The SIP trapezoid



The classic picture of a SIP call is a trapezoid. Each user sends requests to the proxy of its own domain. The two proxies talk to each other. Once the call is set up, later requests and the media can go directly between the two phones.

RFC 3261 §4, Figure 1

1.7 Request and response



SIP works like HTTP. A request names a method and a target. The answer is a response with a three digit status code. A request, all its provisional responses and its final response form one transaction. A whole call consists of several transactions.

RFC 3261 §7, §17

1.8 Three things to remember

1

SIP + SDP + RTP

Signalling, description and media are separate.

2

Address of record and contact

Registration binds the stable name to the current device.

3

Text, like HTTP

Methods, headers, status codes.

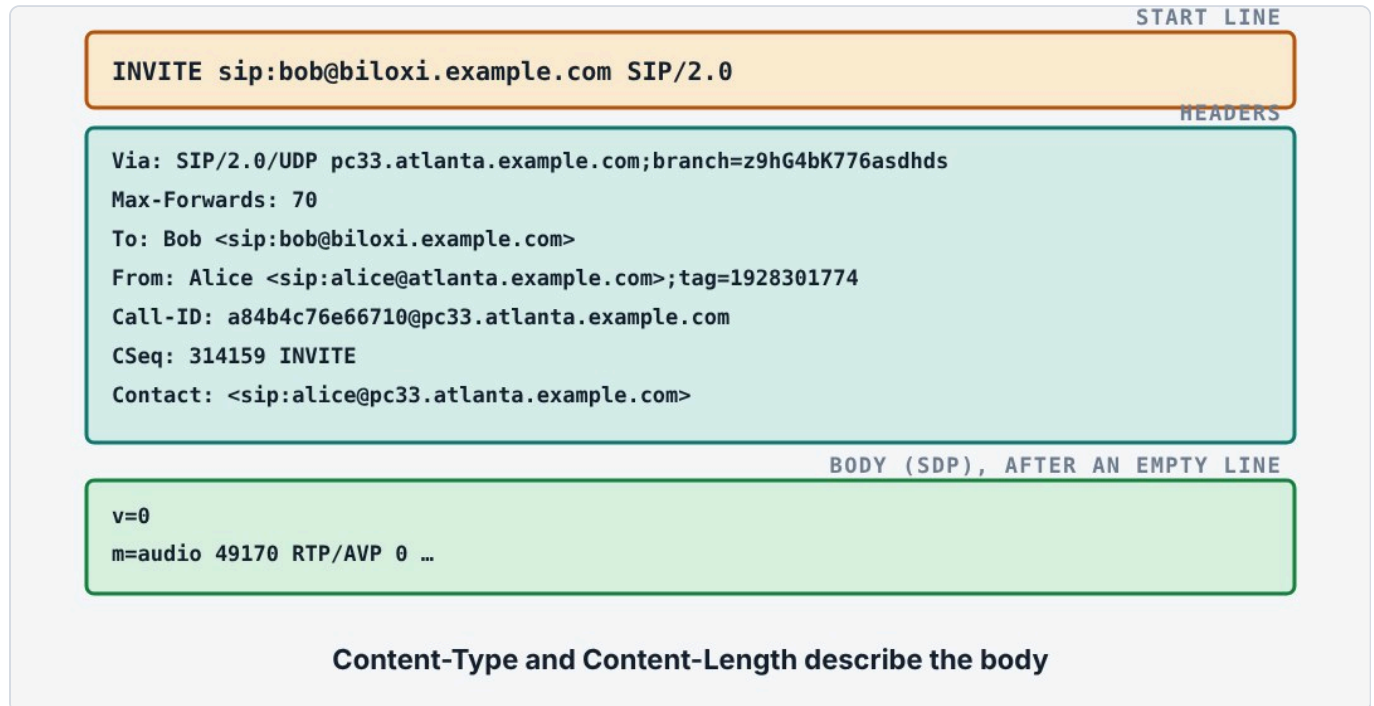
SIP sets up sessions, SDP describes them, RTP carries them. Users are reached through proxies and a registrar. Everything is request and response.

RFC 3261

SIP Message Structure

How to read a SIP message: start line, headers, body, and the methods.

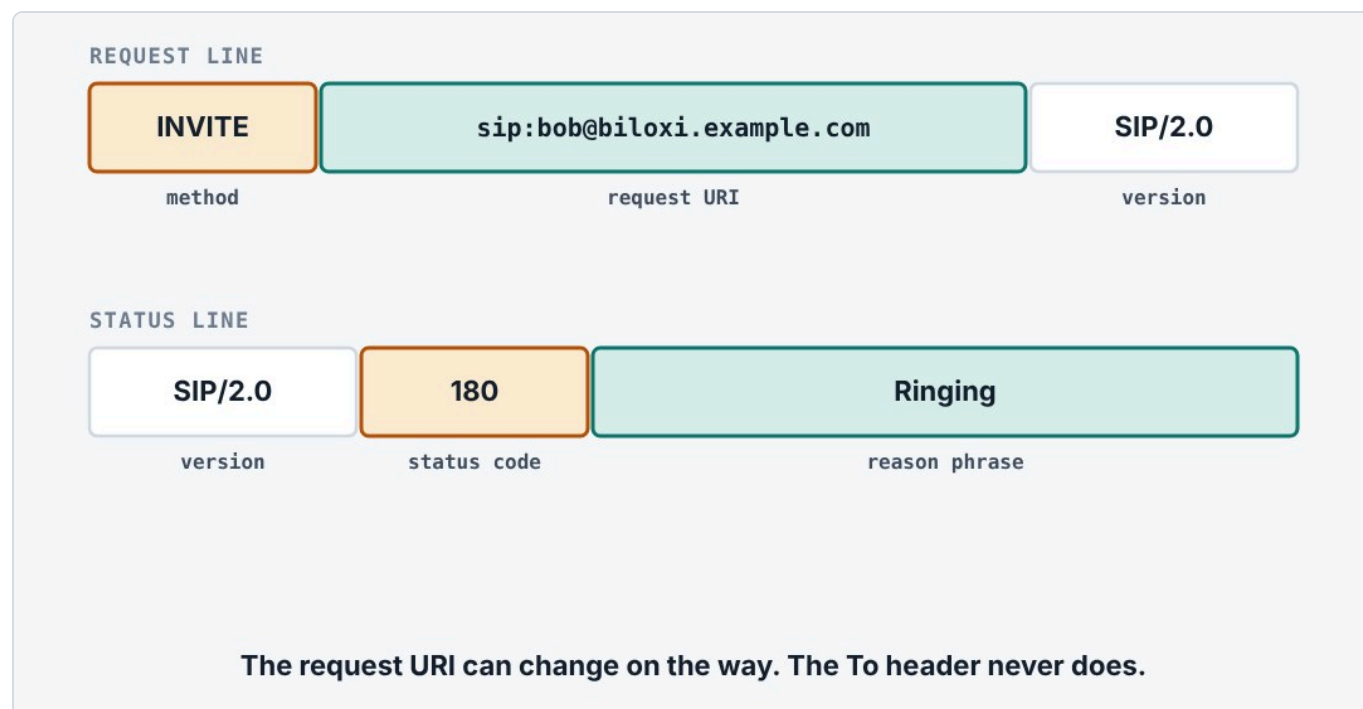
2.1 Anatomy of a message



Every SIP message has three parts. The start line says what the message is. The headers carry routing and identification. After an empty line the body follows, most often SDP. Requests and responses differ only in the start line.

RFC 3261 §7

2.2 Request line and status line



A request starts with the method, the request URI and the version. The request URI is where the message should go now, and proxies may rewrite it. A response starts with the version, a status code and a reason phrase. Only the code matters, the phrase is for humans.

RFC 3261 §7.1, §7.2

2.3 The six headers every request has

Via	path so far. Responses follow it back. branch = transaction.
From	who started the request, with a tag
To	the logical recipient. The answering side adds its tag.
Call-ID	one random identifier for the whole call
CSeq	sequence number and method. Orders requests in a dialog.
Max-Forwards	hop counter, starts at 70, stops loops

Six headers are mandatory in every request. Together they identify the transaction and the dialog and prevent loops. When you open a trace, these are the first ones to look at.

RFC 3261 §8.1.1

2.4 Headers for routing

ROUTING HEADERS

Contact	direct address of the user agent, target for later requests
Record-Route	added by a proxy to stay on the path of the dialog
Route	the list of proxies a request must pass
Path	like Record-Route, but for REGISTER (IMS: P-CSCF)
Service-Route	returned in 200 to REGISTER: route for later requests (IMS: S-CSCF)

A second group of headers controls where later messages go. Contact gives the direct address of the sender. A proxy that wants to stay in the path adds itself to Record-Route. The endpoints copy that list into Route headers of every later request in the dialog.

RFC 3261 §8.1.1.8, §16.6, §20.30, §20.34

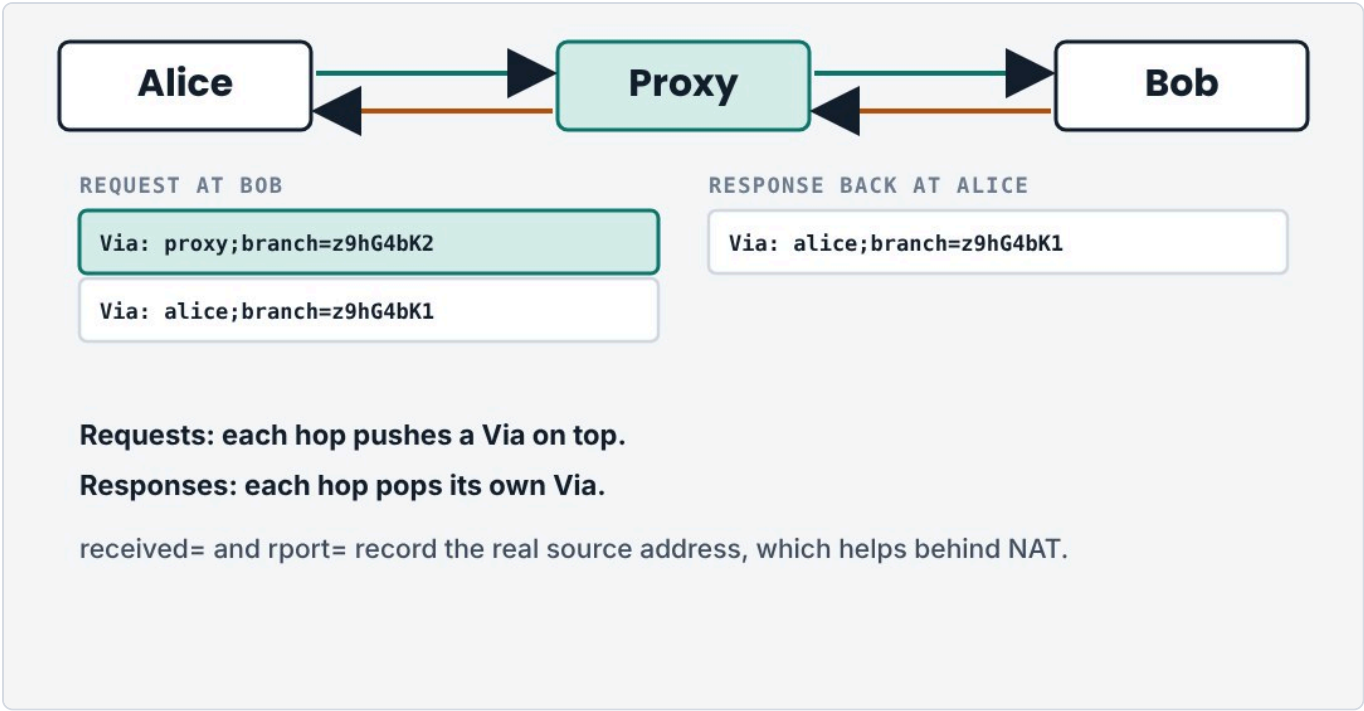
2.5 The methods

METHOD	PURPOSE	DEFINED IN
INVITE	start a session, or change it (re-INVITE)	RFC 3261
ACK	confirm the final response to INVITE	RFC 3261
BYE	end a session	RFC 3261
CANCEL	stop a request that has no final answer yet	RFC 3261
REGISTER	bind address of record to contact	RFC 3261
OPTIONS	ask for capabilities, also used as keep-alive	RFC 3261
PRACK	acknowledge a reliable provisional response	RFC 3262
UPDATE	change session parameters before the answer	RFC 3311
SUBSCRIBE, NOTIFY	ask for and deliver event state	RFC 6665
REFER, MESSAGE	call transfer, instant message or SMS	RFC 3515, 3428

The base specification defines six methods. Extensions added more. In a VoLTE trace you will meet almost all of them. INVITE is the only method that is answered with a three-way handshake.

RFC 3261 · RFC 3262 · RFC 3311 · RFC 6665 · RFC 3515 · RFC 3428

2.6 How Via and branch work



Each hop adds its own Via on top before it forwards a request. The response carries the same list, and each hop removes its own entry and sends the response to the next one. The branch parameter starts with z9hG4bK and identifies the transaction at that hop.

RFC 3261 §8.1.1.7, §18.2.2

2.7 Compact forms and P-headers

SHORT	HEADER	NOTE
v	Via	
f, t	From, To	
i	Call-ID	
m	Contact	
l, c	Content-Length, Content-Type	
IMS HEADER		CARRIES
P-Asserted-Identity		caller identity, verified by the network
P-Preferred-Identity		identity the UE would like to use
P-Access-Network-Info		radio access type and cell identity
P-Charging-Vector		charging correlation identifier (ICID)

Header names have one-letter short forms, used to keep messages small. IMS adds its own private headers, which carry what the mobile network knows: the access network, the asserted identity of the caller and charging identifiers.

RFC 3261 §7.3.3 · RFC 7315 · RFC 3325

2.8 Three things to remember

1

Read the start line first

Method and request URI, or status code.

2

Call-ID, From tag, To tag

The three values that tie a call together in a trace.

3

Via back, Route forward

Responses follow Via. Later requests follow the Route set.

Start line, headers, body. Call-ID plus the two tags identify the dialog. Via brings the response back.

RFC 3261 §7, §8

Responses, Transactions and Dialogs

The six response classes, how a transaction works, the timers behind retransmissions, and what a dialog is.

3.1 Six classes of responses

CLASS	MEANING	EXAMPLES
1xx	provisional	100 Trying, 180 Ringing, 183 Session Progress
2xx	success	200 OK, 202 Accepted
3xx	redirection	301 Moved Permanently, 302 Moved Temporarily
4xx	client error	401, 403, 404, 408, 480, 486, 487, 488
5xx	server error	500, 503 Service Unavailable, 504
6xx	global failure	603 Decline, 604 Does Not Exist Anywhere

The first digit of the status code gives the class. 1xx responses are provisional: the request is in progress. All others are final and end the transaction. 2xx means success. 3xx redirects. 4xx is an error on the client side, 5xx on the server side, and 6xx a failure that no other server can fix.

RFC 3261 §21

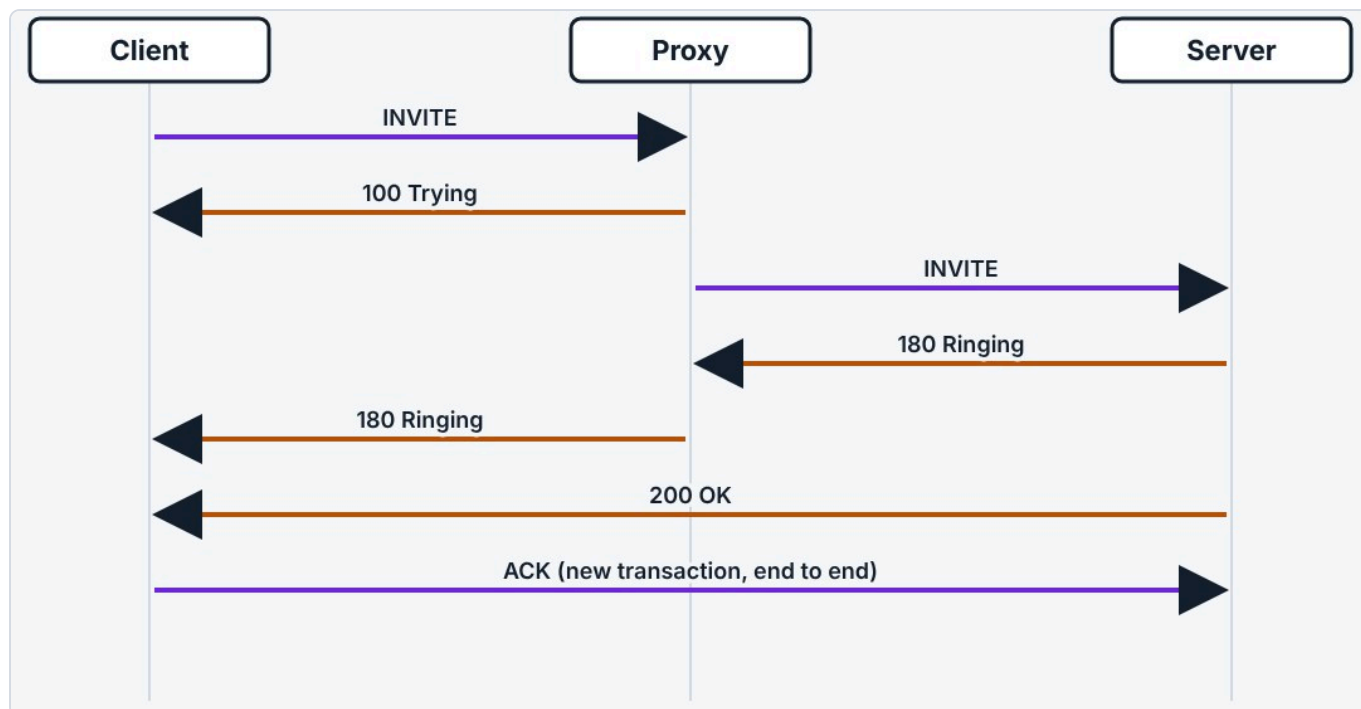
3.2 The codes you meet most

CODE	PHRASE	USUAL MEANING
401	Unauthorized	registrar asks for credentials
403	Forbidden	not allowed: barring, wrong identity
404	Not Found	user does not exist in this domain
408	Request Timeout	no answer in time from the next hop
480	Temporarily Unavailable	user not registered or not reachable
486	Busy Here	user is busy
487	Request Terminated	INVITE was cancelled
488	Not Acceptable Here	no common codec or media
503	Service Unavailable	overload, node down, bearer failure
504	Server Time-out	a server further on did not answer

A small set of codes explains most failed calls. 401 and 407 are not errors but challenges. 486 means the user is busy and 480 that the user is not reachable. 487 is the normal answer to an INVITE that was cancelled. 488 points at the SDP, and 503 at an overloaded or broken server.

RFC 3261 §21.4, §21.5

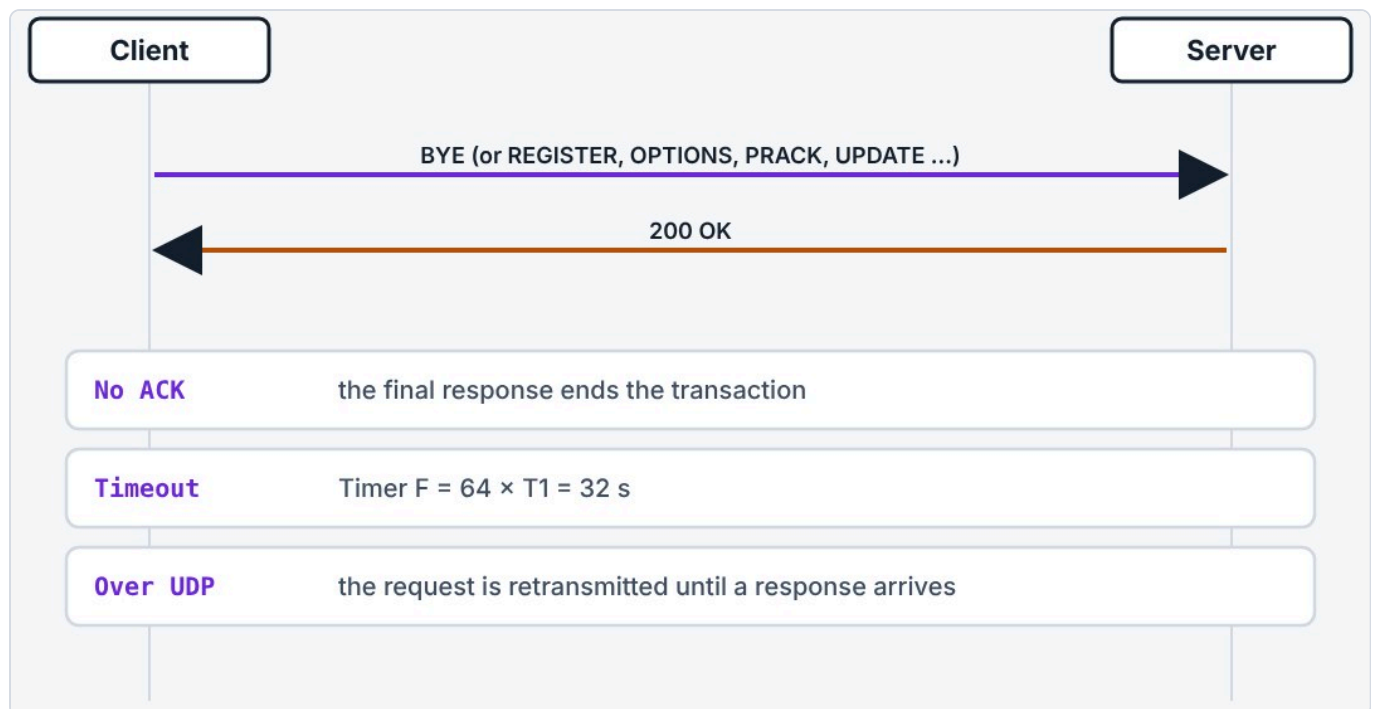
3.3 The INVITE transaction



An INVITE can take a long time, because a human has to answer. It therefore gets a three-way handshake. The server sends 100 Trying at once to stop retransmissions. A final response is retransmitted until the ACK arrives. For a 2xx, the ACK is a separate transaction sent end to end.

RFC 3261 §17.1.1, §13.2.2.4

3.4 Non-INVITE transactions



All other methods use a simple two-way exchange: one request, one final response, no ACK. They are expected to finish quickly. If no final response arrives within 32 seconds the transaction fails, and the client treats it like a 408.

RFC 3261 §17.1.2

3.5 Timers and retransmission



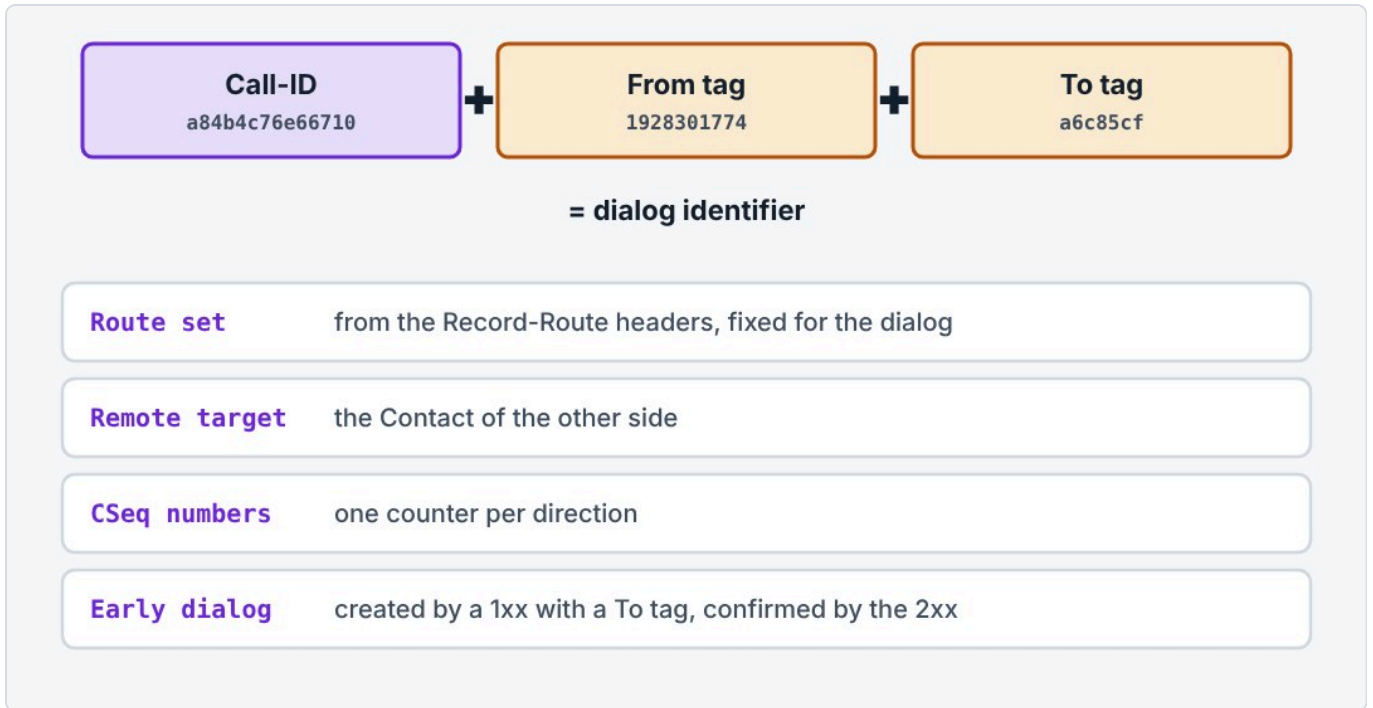
TIMER	VALUE	MEANING
T1	500 ms	round trip estimate, first retransmission
T2	4 s	largest interval for non-INVITE retransmissions
Timer A	T1, doubling	INVITE retransmission interval
Timer B	$64 \times T1 = 32 \text{ s}$	INVITE transaction timeout
Timer F	$64 \times T1 = 32 \text{ s}$	non-INVITE transaction timeout

IMS: $T1 = 2 \text{ s}$ on the radio side, so Timer B and F become 128 s

Over UDP, SIP must repeat messages itself. The basic timer T1 is 500 ms, an estimate of the round trip time. A request is retransmitted after T1, then after twice that, and so on. After 64 times T1, which is 32 seconds, the transaction gives up. Mobile networks often use longer values.

RFC 3261 §17.1.1.2, Table 4 · TS 24.229 Table 7.7.1

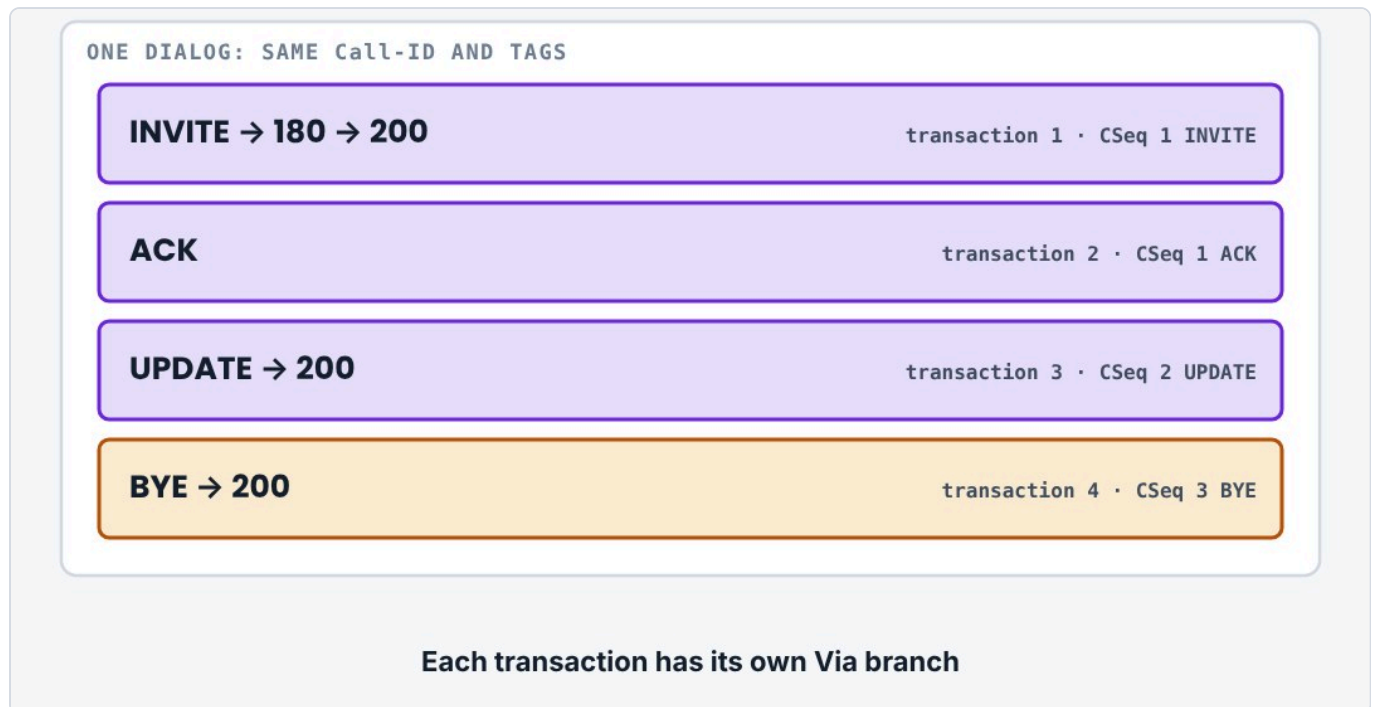
3.6 What a dialog is



A dialog is the lasting relation between two user agents, for example for the length of a call. It is identified by three values: the Call-ID, the local tag and the remote tag. It also stores the route set, the remote target and the sequence numbers. INVITE and SUBSCRIBE create dialogs.

RFC 3261 §12

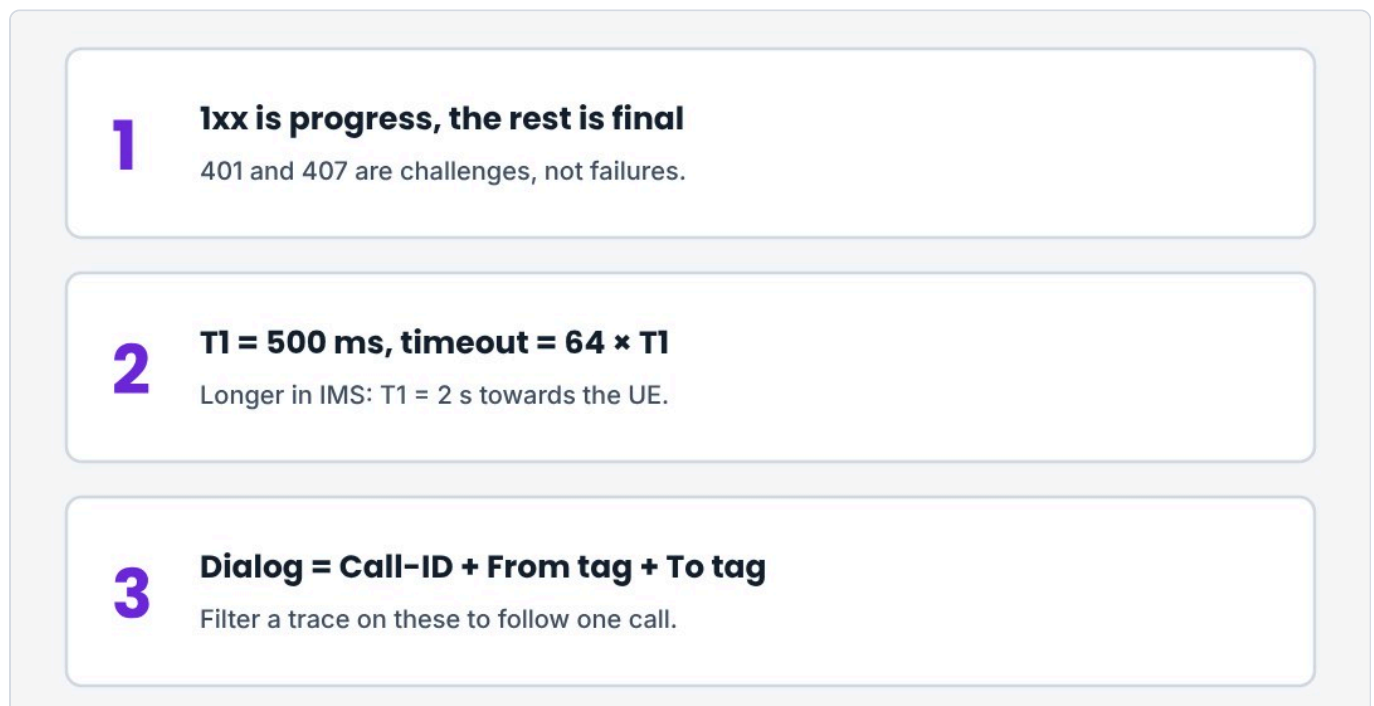
3.7 Transactions inside a dialog



A call is one dialog made of several transactions. The INVITE transaction creates it. The ACK is its own transaction. Later a re-INVITE or UPDATE may change the session. The BYE transaction ends the dialog. All of them share the Call-ID and the tags, and each has its own branch and CSeq.

RFC 3261 §12, §17

3.8 Three things to remember



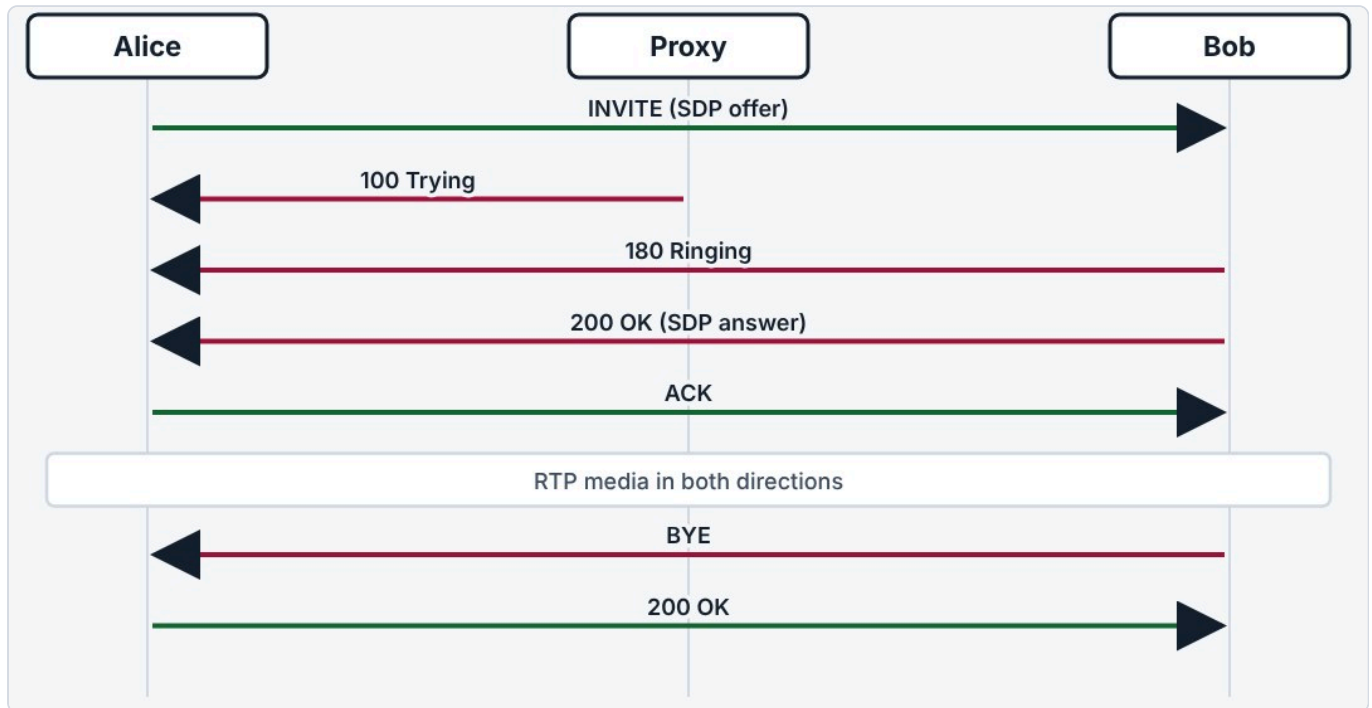
The first digit of the code tells the story. INVITE has a three-way handshake, everything else two. A dialog is Call-ID plus two tags.

RFC 3261 §12, §17, §21

The Basic Call Flow

A complete call from INVITE to BYE, and the variations: cancel, busy, reliable provisional responses and re-INVITE.

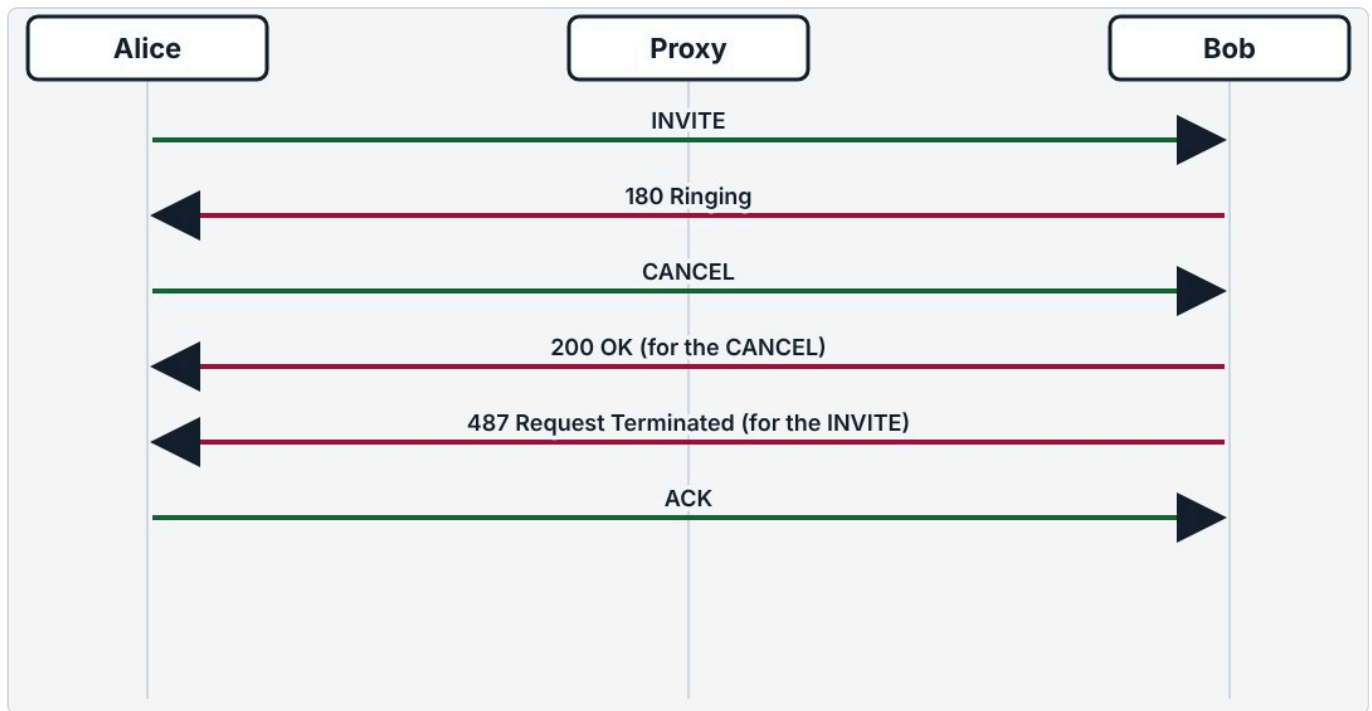
4.1 A successful call



This is the flow behind every call. The INVITE carries the SDP offer. 100 Trying stops retransmissions, 180 Ringing tells the caller that the phone rings. The 200 OK carries the SDP answer when the user picks up. The ACK completes the setup, the media flows, and a BYE from either side ends it.

RFC 3261 §4, §13 · RFC 3665 §3.1

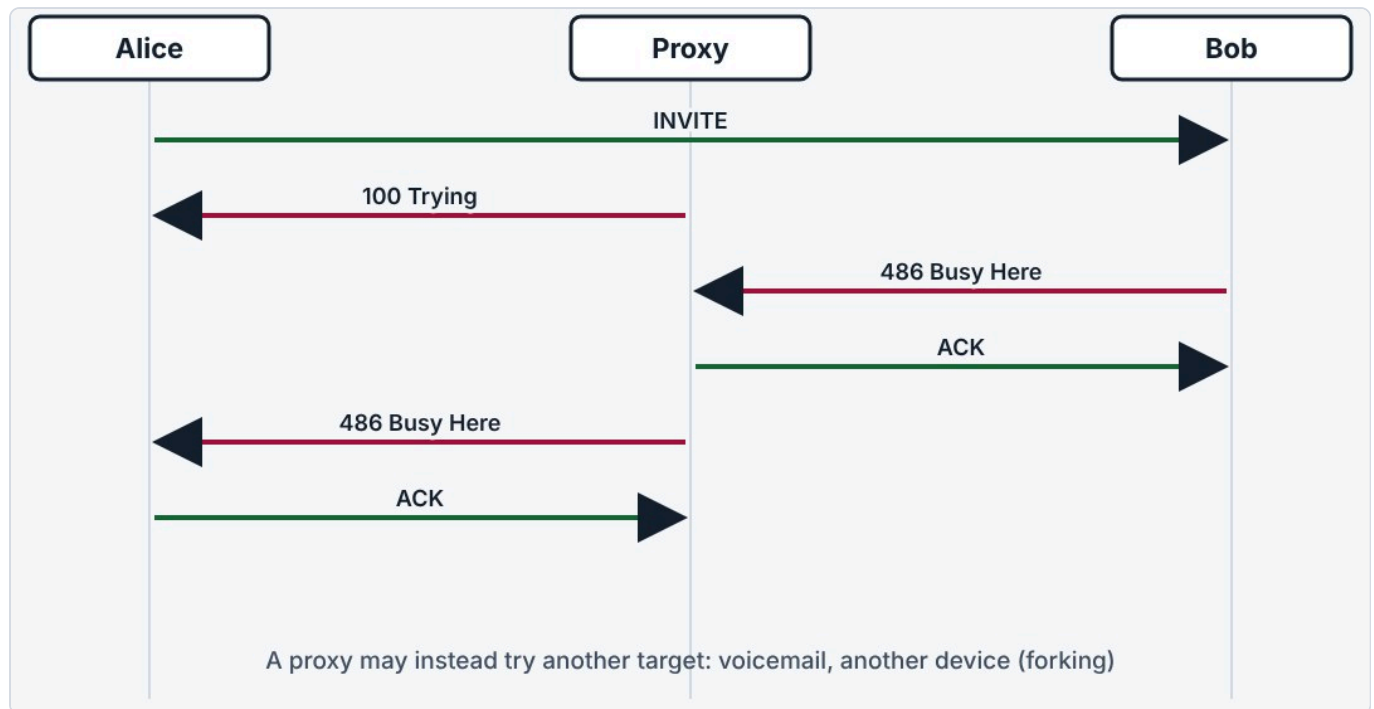
4.2 The caller gives up: CANCEL



If the caller hangs up while it still rings, it sends CANCEL. CANCEL is answered hop by hop with 200 OK. The called side then ends the original INVITE with 487 Request Terminated, which the caller acknowledges. CANCEL only works before a final response. After the answer, a BYE is needed.

RFC 3261 §9

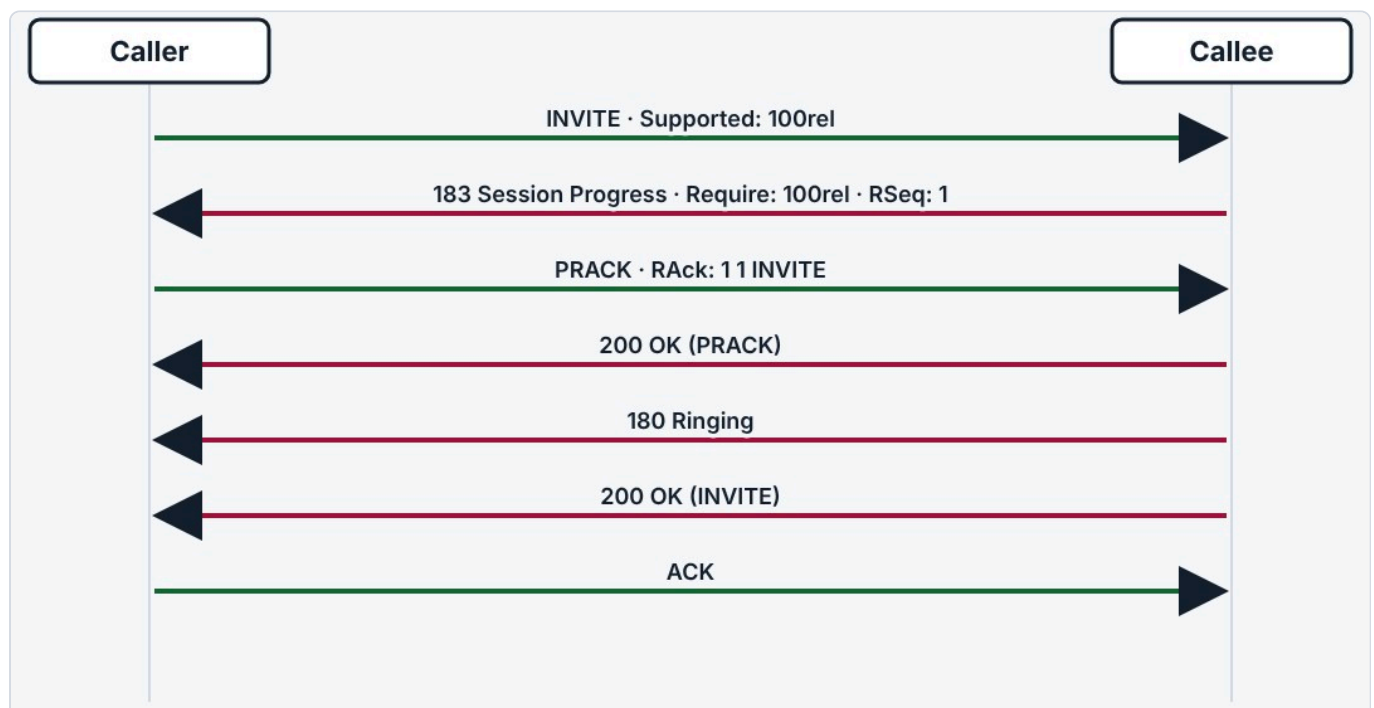
4.3 The call is rejected



When the called party is busy or declines, the INVITE gets a final error response. A final response to INVITE is always acknowledged with ACK, also when it is an error. For non-2xx responses the ACK is part of the same transaction and is sent hop by hop.

RFC 3261 §17.1.1.3

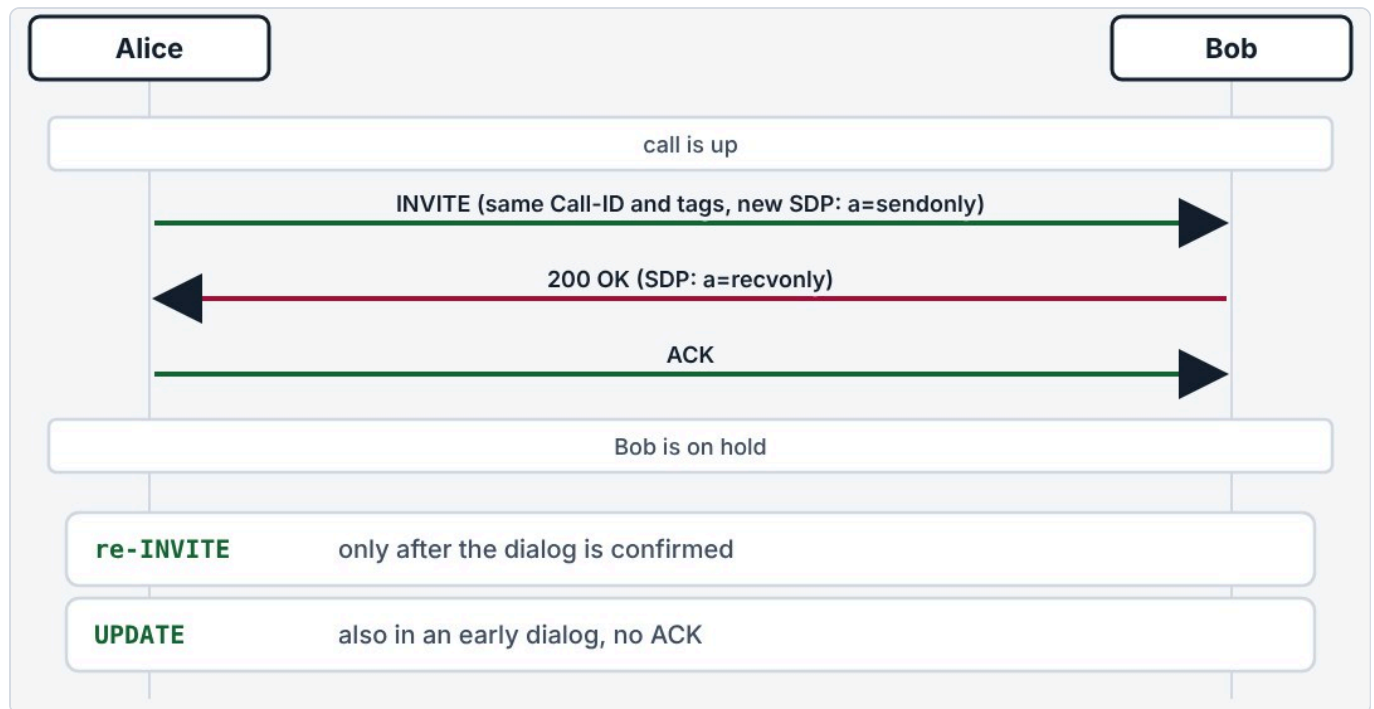
4.4 Reliable provisional responses



Normal 1xx responses can get lost. When a 183 carries SDP that the caller must receive, it is sent reliably. The response carries Require: 100rel and an RSeq number. The caller confirms it with PRACK, which itself gets a 200 OK. VoLTE uses this in every call.

RFC 3262

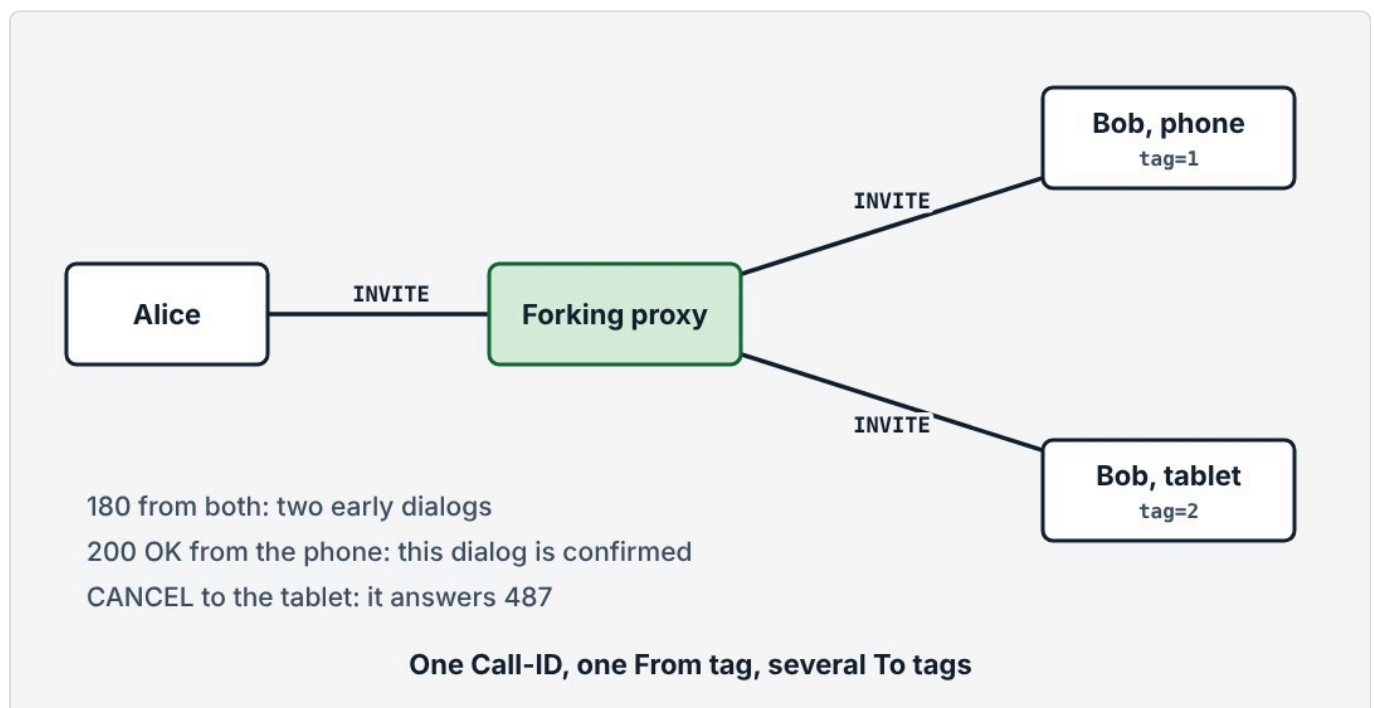
4.5 Changing a call: re-INVITE and UPDATE



A session can change while it runs: hold, resume, add video, change codec. A re-INVITE is an INVITE inside the existing dialog with a new SDP offer. UPDATE does the same but can also be used before the call is answered, which is why VoLTE uses it during setup.

RFC 3261 §14 · RFC 3311

4.6 Forking



A user may be registered on several devices. A forking proxy sends the INVITE to all of them, in parallel or one after the other. Each device answers with its own To tag, so the caller sees several early dialogs. The first 200 OK wins, and the proxy cancels the other branches.

RFC 3261 §16.7, §12.1

4.7 Session timers

SESSION TIMER

Session-Expires	the interval, for example 1800 s, and who refreshes
Min-SE	the shortest interval a node accepts, at least 90 s
refresher	uac or uas: the side that must send the refresh
Refresh	re-INVITE or UPDATE at half the interval
422	Session Interval Too Small: retry with the Min-SE value

A proxy cannot know whether a call is still alive if the BYE is lost. Session timers solve this. The endpoints agree on an interval in the Session-Expires header, and one of them refreshes the session with a re-INVITE or UPDATE before it runs out. Without a refresh the session is considered dead.

RFC 4028

4.8 Three things to remember

- 1 Offer in INVITE, answer in 200**
Or earlier, in a reliable 183.
- 2 CANCEL → 200, then 487 → ACK**
Two transactions end the unanswered call.
- 3 re-INVITE and UPDATE**
Same dialog, new SDP: hold, resume, codec change.

INVITE, 180, 200, ACK, then BYE. CANCEL before the answer, BYE after it. Every final response to INVITE gets an ACK.

RFC 3261 §9, §13, §15

SDP and Offer/Answer

How two endpoints agree on codecs, addresses and ports, and how to read the SDP in a trace.

5.1 What SDP describes

```
v=0
o=alice 2890844526 2890844526 IN IP4 192.0.2.10
s=-
c=IN IP4 192.0.2.10
t=0 0
m=audio 49170 RTP/AVP 96 97 8
a=rtpmap:96 AMR-WB/16000
a=fmtp:96 mode-change-capability=2
a=rtpmap:97 telephone-event/16000
a=rtpmap:8 PCMA/8000
a=ptime:20
a=sendrecv
```

v, o, s, t
session level

c
where to send media

m
one media stream

a
attributes

Session level first, then one block per media stream

SDP is a description, not a protocol. It lists the media streams of a session: the type, the transport address where the sender wants to receive, and the formats it supports. Each line is one letter, an equals sign and a value. The order of the lines is fixed.

RFC 8866 §5

5.2 The lines that matter

c=IN IP4 192.0.2.10

connection address: send the RTP here

m=audio 49170 RTP/AVP 96 8

type, port, protocol, payload types by preference

a=rtpmap:96 AMR-WB/16000

payload type 96 means AMR-WB at 16 kHz

a=fmtp:96 ...

codec parameters, for example the allowed modes

a=ptime:20

20 ms of audio per packet

b=AS:41

bandwidth in kbit/s, used for the QoS bearer

A few lines carry almost everything you need in a trace. The c line gives the IP address for media. The m line gives the media type, the port and the list of payload types in order of preference. The rtpmap attributes translate dynamic payload types into codec names.

RFC 8866 §5.7, §5.14, §6.6

5.3 Offer and answer

Offer (in the INVITE)

m=audio 49170 RTP/AVP 96 97 8
a=rtpmap:96 AMR-WB/16000
a=rtpmap:97 AMR/8000
a=rtpmap:8 PCMA/8000



Answer (in 183 or 200)

m=audio 3456 RTP/AVP 96
a=rtpmap:96 AMR-WB/16000

Result: AMR-WB, payload type 96, in both directions

Alice receives on port 49170, Bob on port 3456

The answerer picks from the offer and cannot add new codecs

One side sends an offer with everything it supports. The other side answers with the subset it accepts, keeping the same number of media lines in the same order. After the answer both know which codec to send and where. A stream is refused by setting its port to zero.

RFC 3264 §5, §6

5.4 Where offer and answer can travel

OFFER IN	ANSWER IN
INVITE	reliable 183 or 180, or 200 OK
200 OK (INVITE had no SDP)	ACK
reliable 1xx	PRACK
PRACK	200 OK (PRACK)
UPDATE	200 OK (UPDATE)

"Late offer": an INVITE without SDP, the offer comes in the 200 OK

The offer does not have to be in the INVITE. Several combinations are allowed, as long as each offer gets exactly one answer before the next offer. VoLTE uses the first and, for the second round with the preconditions, PRACK or UPDATE.

RFC 3264 · RFC 6337 §2, Table 1

5.5 Direction attributes and hold

OFFER	ANSWER	MEANING
a=sendrecv	a=sendrecv	normal two-way call
a=sendonly	a=recvonly	offerer holds the call, may play music
a=recvonly	a=sendonly	offerer only listens
a=inactive	a=inactive	no media in either direction

Older phones signal hold with c=IN IP4 0.0.0.0. Do not use it.

Four attributes set the direction of a stream. The default is sendrecv. To put a call on hold, the holding side offers sendonly, and the other side answers recvonly. inactive stops both directions. The same attributes switch early media on and off.

RFC 3264 §5.1, §8.4 · RFC 8866 §6.7

5.6 Preconditions

QoS PRECONDITION ATTRIBUTES

a=curr:qos local none

current: nothing reserved on my side yet

a=curr:qos remote none

and none on the other side, as far as I know

a=des:qos mandatory local sendrecv

desired: QoS must exist before alerting

a=des:qos mandatory remote sendrecv

the same for the remote side

a=conf:qos remote sendrecv

please tell me when your side is ready

When the bearer is up: UPDATE with a=curr:qos local sendrecv

Requires the option tags 100rel and precondition

In a mobile network the phone should not ring before the voice bearer exists. Preconditions solve this. Both sides state in SDP that QoS is required and whether they have it yet. The called phone only starts ringing when both sides have confirmed their resources.

RFC 3312 · RFC 4032 · TS 24.229 §5.1.3, §6.1.2

5.7 Codecs in mobile voice

CODEC	AUDIO BAND	BIT RATES	SAMPLING
AMR	300 – 3400 Hz	4.75 – 12.2 kbit/s	8 kHz
AMR-WB	50 – 7000 Hz	6.6 – 23.85 kbit/s	16 kHz
EVS	up to 20 kHz	5.9 – 128 kbit/s	8 to 48 kHz
PCMA / PCMU	300 – 3400 Hz	64 kbit/s	8 kHz
telephone-event	DTMF digits	–	as the codec

The codec decides the quality and the bit rate. AMR is the narrowband codec inherited from 3G. AMR-WB doubles the audio bandwidth and is known as HD voice. EVS goes further and is more robust to packet loss. DTMF tones are sent as telephone-event packets, not as audio.

TS 26.114 §5.2 · RFC 4867 · RFC 4733 · GSMA IR.92

5.8 Three things to remember

1

Read c=, m= and a=rtpmap

Address, port and codec of each stream.

2

The answer is a subset of the offer

Same media lines, same order.

3

488 means the SDP did not match

No common codec, or an unacceptable media line.

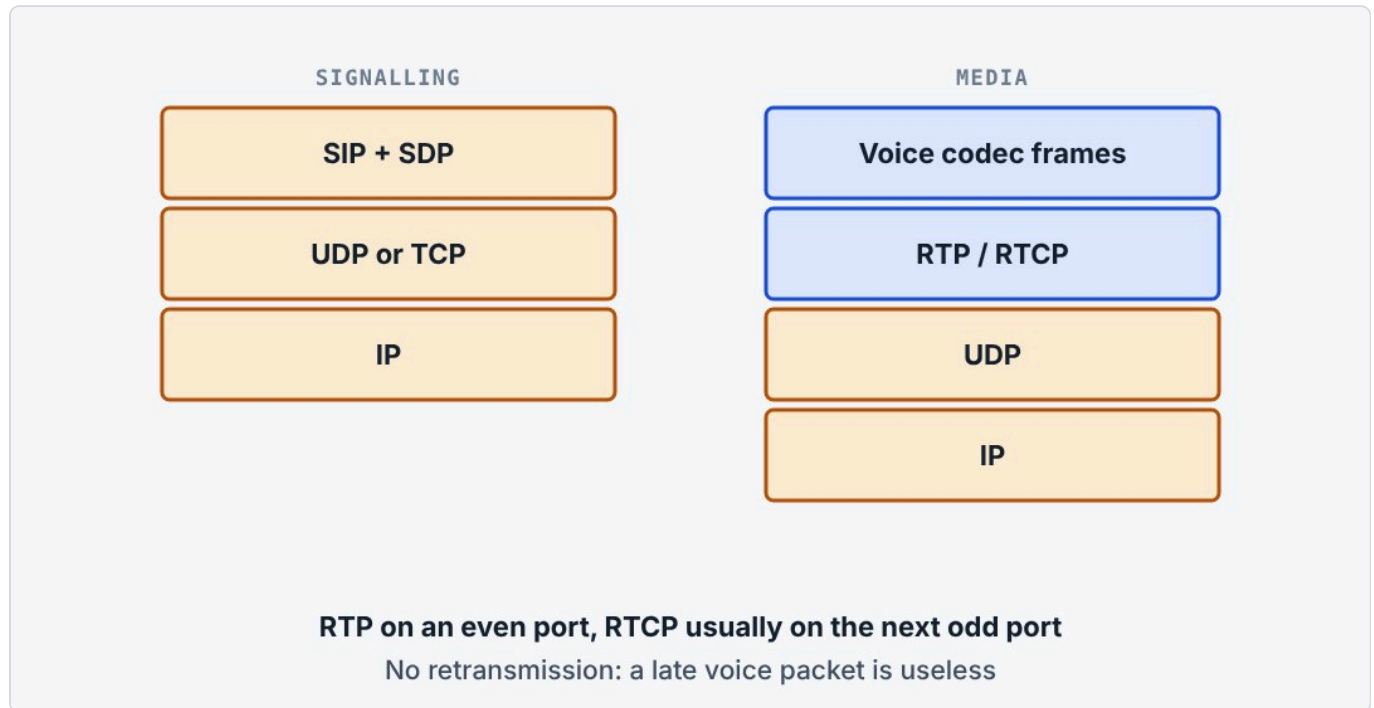
SDP says where and how to send media. One offer, one answer. Direction attributes handle hold, preconditions hold back the ringing.

RFC 8866 · RFC 3264 · RFC 3312

RTP and RTCP

How the voice travels: the RTP header, payload types, jitter, and the quality reports of RTCP.

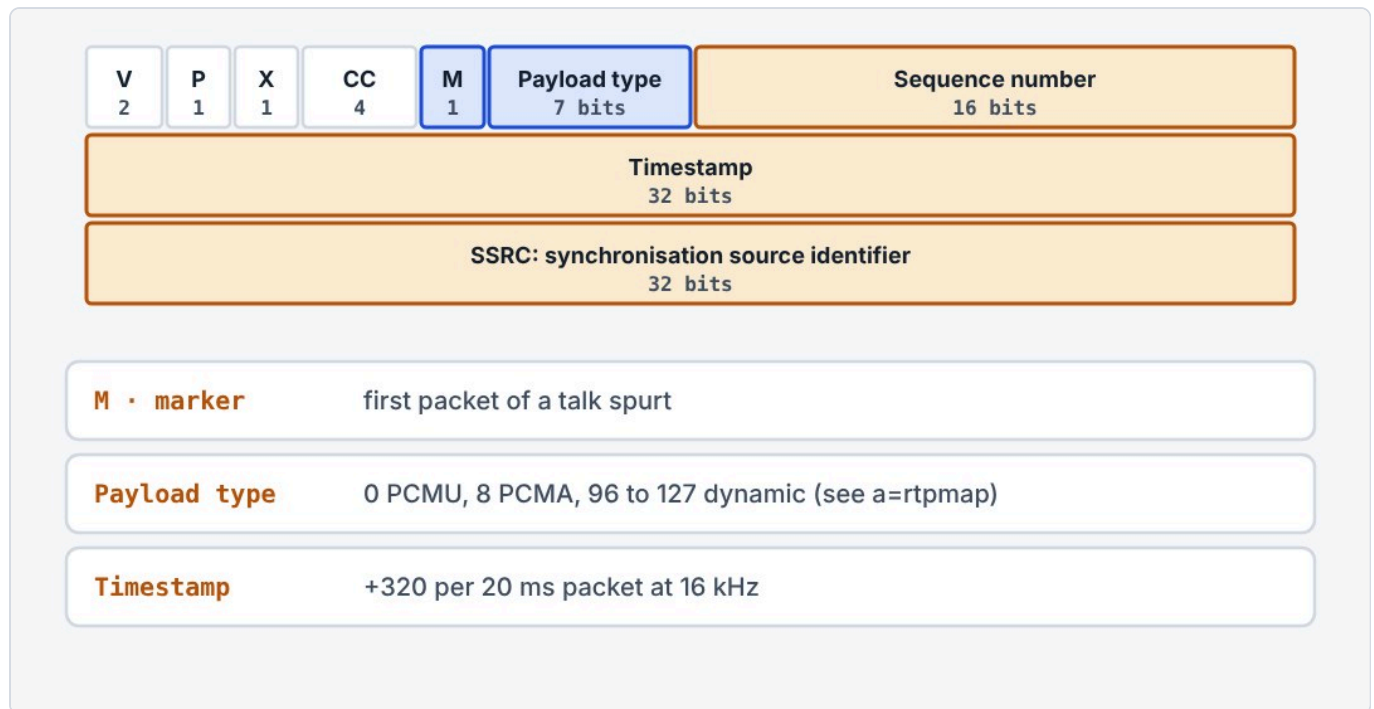
6.1 Media runs beside the signalling



Once SDP has been exchanged, the endpoints send media as RTP over UDP to the address and port from the SDP. RTP adds what UDP lacks for real-time media: a sequence number, a timestamp and a source identifier. RTCP runs next to it and reports on quality.

RFC 3550 §1, §2

6.2 The RTP header

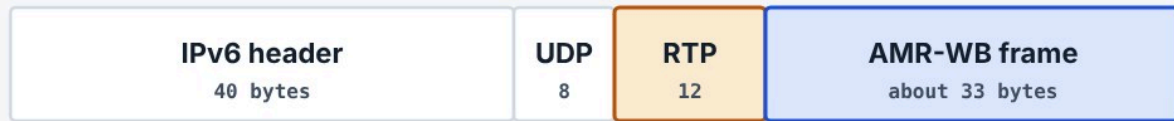


The fixed RTP header is 12 bytes. The payload type says which codec the packet carries. The sequence number rises by one per packet and reveals loss and reordering. The timestamp counts sampling instants and lets the receiver play the audio at the right time. The SSRC names the source.

RFC 3550 §5.1

6.3 Packets on the wire

ONE VOICE PACKET, IPv6



AFTER HEADER COMPRESSION (ROHC), ON THE RADIO



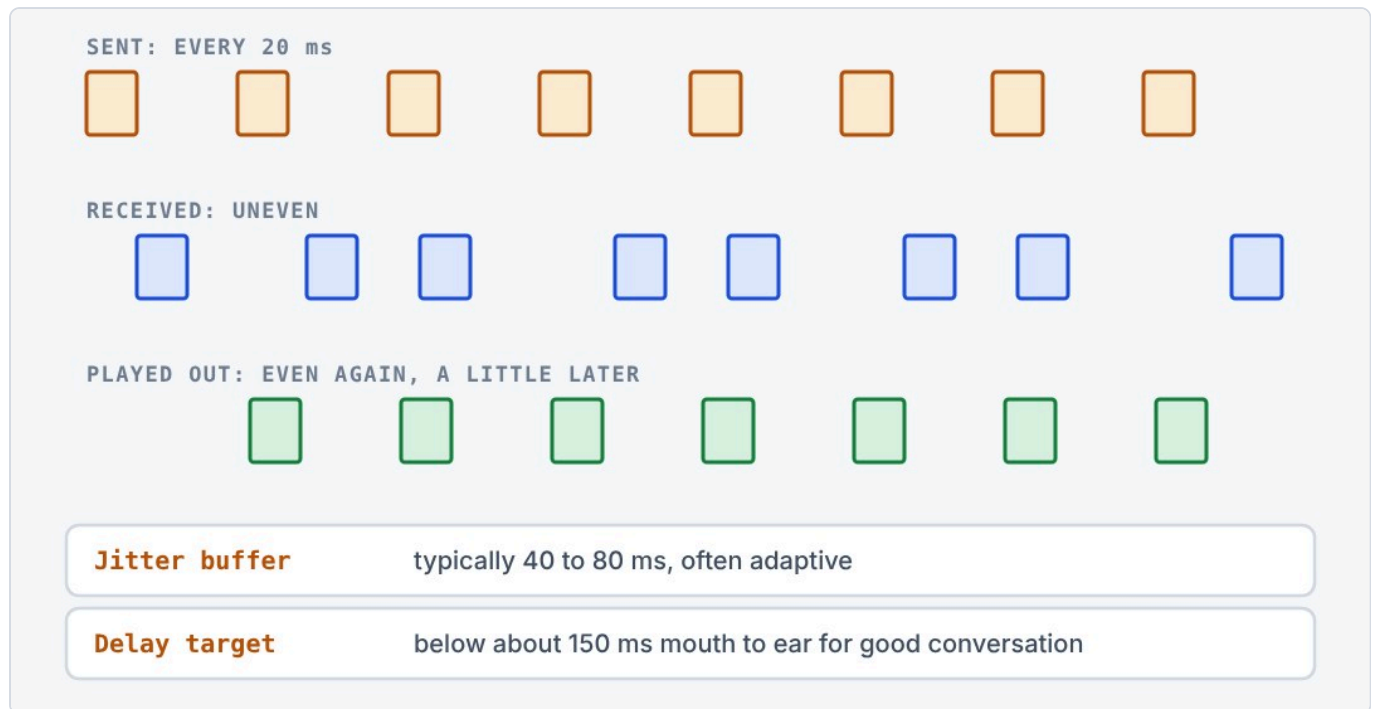
50 packets per second, each way

Without compression: about 37 kbit/s on IPv6. With ROHC: about 15 kbit/s.

A voice call sends one packet every 20 ms in each direction. With AMR-WB at 12.65 kbit/s the payload is about 33 bytes. The headers of IP, UDP and RTP add 40 bytes with IPv4 or 60 with IPv6, more than the voice itself. This is why VoLTE compresses the headers on the radio.

RFC 4867 · RFC 3095 (ROHC) · TS 26.114

6.4 Jitter and the playout buffer



Packets leave every 20 ms but do not arrive every 20 ms. The variation is jitter. The receiver holds packets in a small buffer and plays them out at a steady pace. A bigger buffer hides more jitter but adds delay. Packets that come too late are dropped and concealed by the codec.

RFC 3550 §6.4.1, Appendix A.8

6.5 RTCP: the feedback channel

TYPE	NAME	CONTENT
SR	sender report	packets and bytes sent, NTP and RTP timestamp
RR	receiver report	fraction lost, total lost, jitter, last SR, delay
SDES	source description	CNAME: stable name of the source
BYE	goodbye	the source leaves
APP	application	application-specific data

RTCP share: about 5 % of the session bandwidth (b=RS, b=RR in SDP)

RTCP packets travel alongside RTP every few seconds. A sender report gives the packet and byte counts and links the RTP timestamp to wall clock time. A receiver report tells the sender how its stream arrived: fraction lost, cumulative loss, jitter and the round trip time data.

RFC 3550 §6

6.6 DTMF

TELEPHONE-EVENT PAYLOAD

SDP a=rtpmap:97 telephone-event/16000 and a=fmtp:97 0-15

Event 0 to 9 the digits, 10 = *, 11 = #, 12 to 15 = A to D

E bit end of the event, sent three times for safety

Duration in timestamp units, grows with each packet of the event

Same timestamp all packets of one digit share it; the marker is on the first

Other ways: SIP INFO, or in-band audio (unreliable with compressing codecs)

When you press a key during a call, the digit is not sent as an audible tone inside the codec. It is sent as a telephone-event: special RTP packets with their own payload type that name the digit, its volume and its duration. The last packet is sent three times with the end bit set.

RFC 4733

6.7 What bad voice looks like in a trace

SYMPTOM	LOOK AT
Choppy voice	sequence gaps (loss), arrival intervals (jitter)
One-way audio	c= and m= of both SDPs, NAT, dedicated bearer
No audio at all	was SDP answered? is RTP sent to the right port?
Silence after hold	direction attributes after the re-INVITE
Robot voice, delay	jitter buffer overrun, late packets, handover

Voice problems show in the RTP stream. Gaps in the sequence numbers mean loss. A changing interval between arrivals means jitter. One-way audio usually means the SDP gave a wrong address or a bearer or firewall blocks one direction. No RTP at all after the 200 OK points to SDP or the bearer.

RFC 3550 · RFC 3611 (RTCP XR)

6.8 Three things to remember

- 1 12 byte header**
Payload type, sequence number, timestamp, SSRC.
- 2 Jitter buffer trades delay for smoothness**
Late packets count as lost.
- 3 RTCP SR and RR**
Loss, jitter and round trip time, every few seconds.

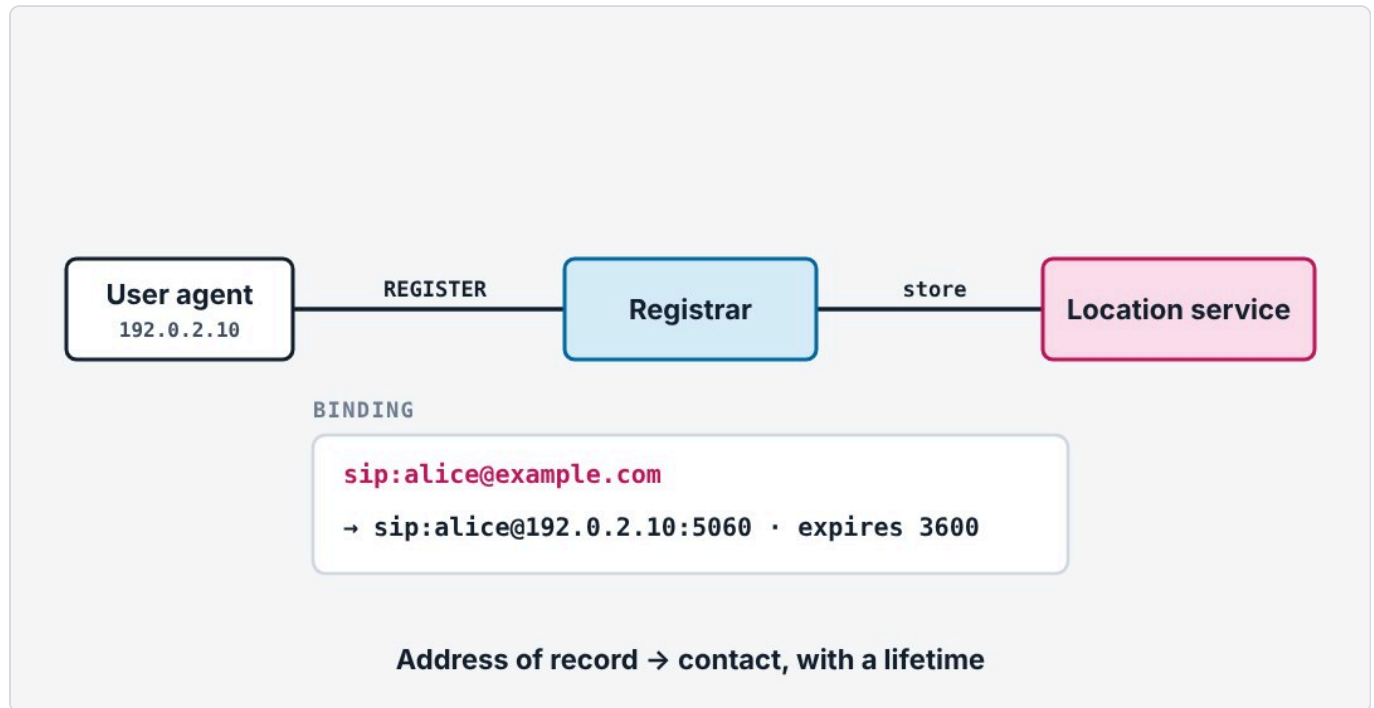
RTP over UDP, one packet per 20 ms. Sequence number for loss, timestamp for timing. RTCP reports the quality.

RFC 3550

Registration and Authentication

How a user agent tells the network where it is, and how the network checks who it is.

7.1 Why register



Calls are addressed to a stable name, but the device moves and its IP address changes. REGISTER tells the registrar: requests for this address of record should go to this contact, for this long. The proxy looks up that binding whenever a call arrives for the user.

RFC 3261 §10

7.2 The REGISTER request

```
REGISTER sip:example.com SIP/2.0
Via: SIP/2.0/UDP 192.0.2.10:5060;branch=z9hG4bKnashds7
To: <sip:alice@example.com>
From: <sip:alice@example.com>;tag=456248
Call-ID: 843817637684230@998sdasdh09
CSeq: 1826 REGISTER
Contact: <sip:alice@192.0.2.10>
Expires: 3600
```

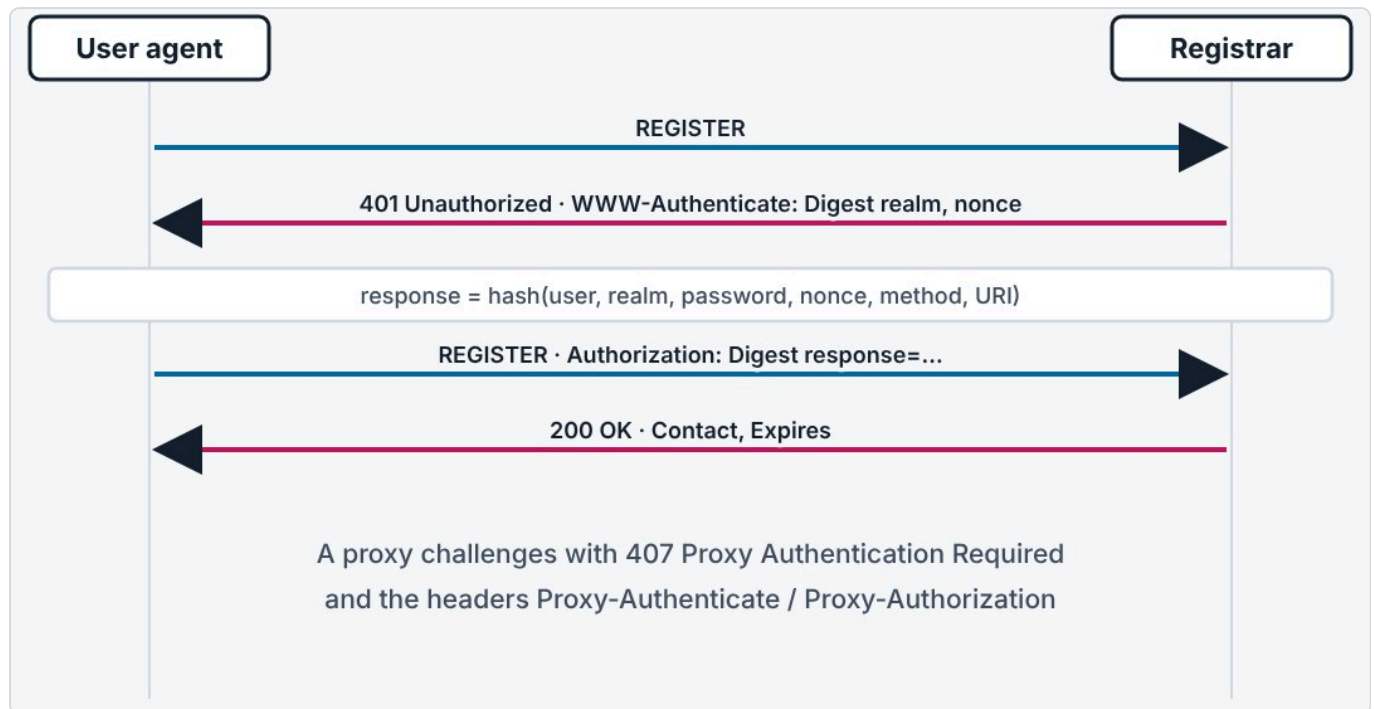
Refresh same Call-ID, higher CSeq, before the binding expires

De-register Expires: 0, or Contact: * with Expires: 0

In a REGISTER the request URI names the domain, not a user. To holds the address of record being registered, From says who registers, and they are usually the same. Contact carries the device address and Expires the lifetime. Call-ID stays the same for all refreshes and CSeq goes up.

RFC 3261 §10.2

7.3 Digest authentication



The registrar must be sure who is registering. With digest authentication it answers the first REGISTER with 401 and a random nonce. The client sends the request again with a response computed from the user name, the password, the nonce and the request. The password itself never crosses the network.

RFC 3261 §22 · RFC 7616

7.4 What the 200 OK tells you

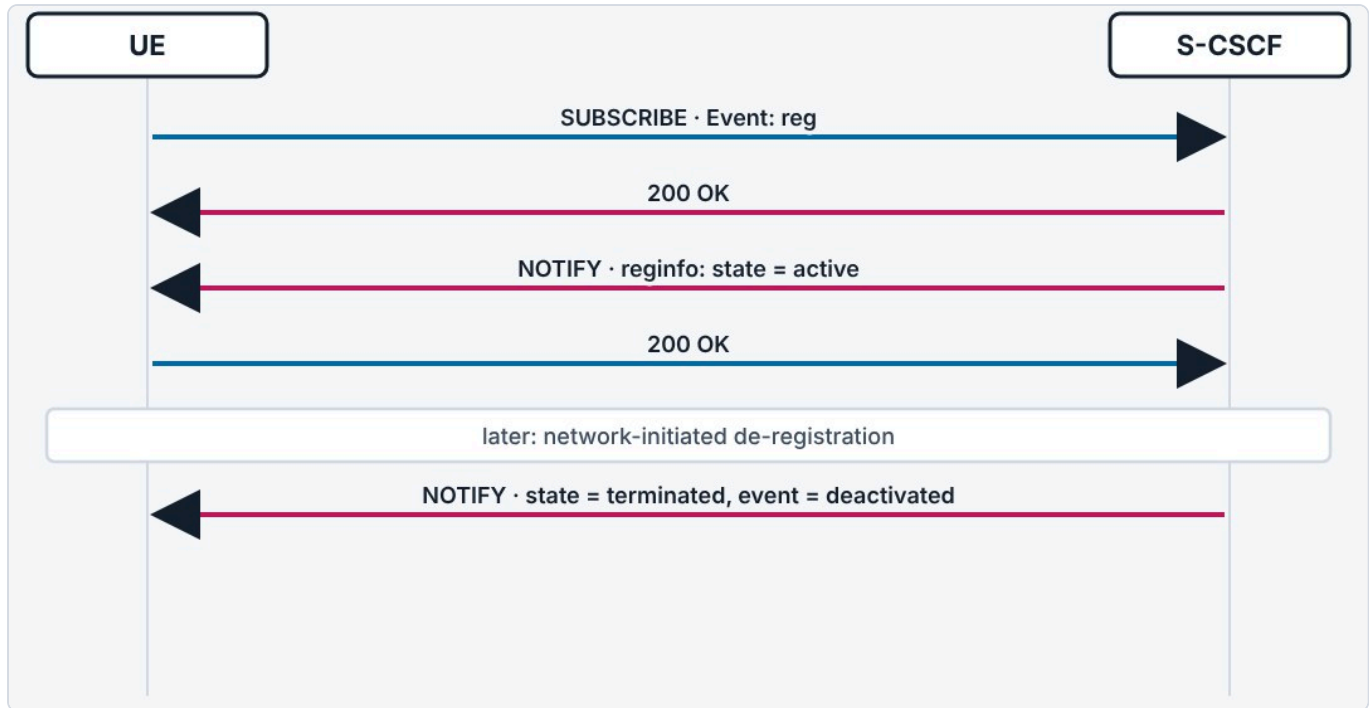
REGISTER · 200 OK

Contact; expires=	the binding and its granted lifetime
Service-Route	IMS: the S-CSCF address, used as Route in later requests
Path	IMS: the P-CSCF, so calls to the UE go through it
P-Associated-URI	IMS: all public identities registered implicitly
423 Interval Too Brief	registrar wants a longer Expires, see Min-Expires

The final 200 OK lists the bindings that now exist and how long each lasts. The registrar may shorten the requested time. In IMS the answer carries more: the route to use for later requests and all public identities that were registered together.

RFC 3261 §10.3 · RFC 3608 · RFC 3327 · RFC 7315

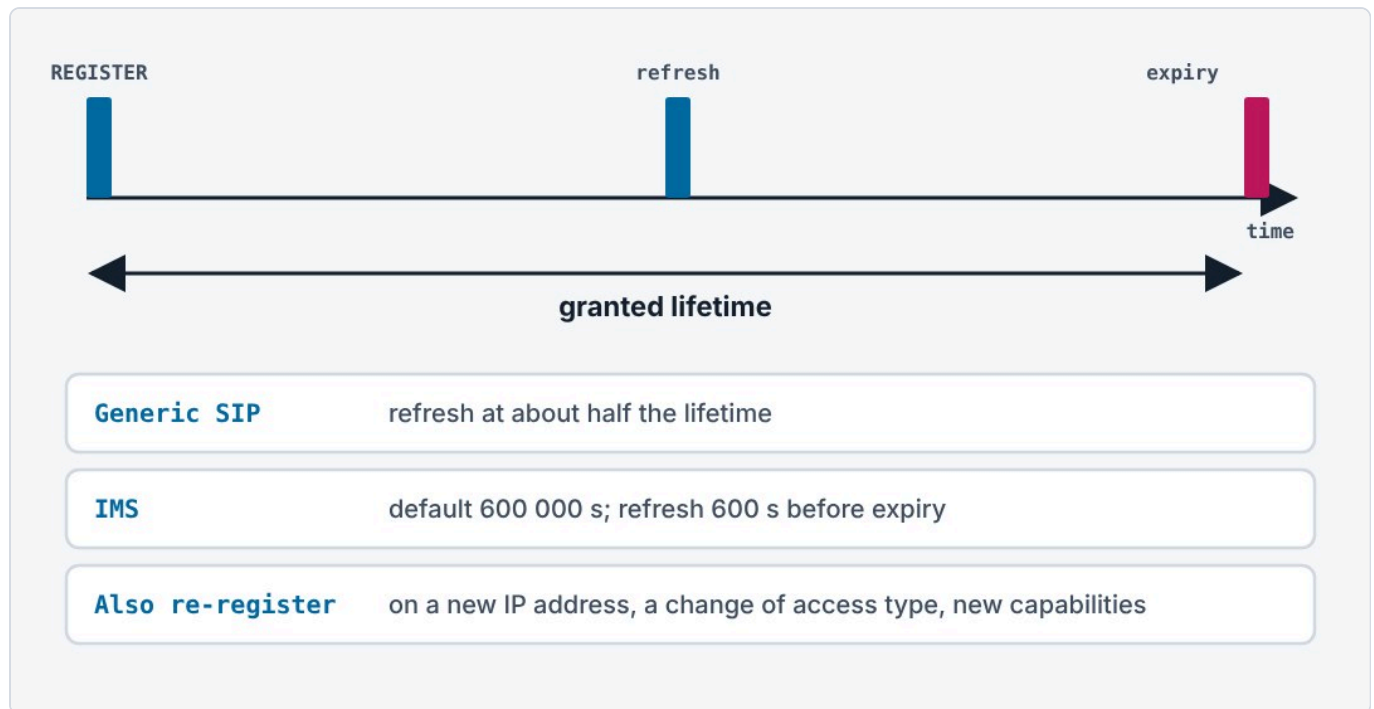
7.5 Registration state: the reg event



A device and the network both need to know when the registration changes, for example when the network removes it. The user agent subscribes to the reg event package. The registrar sends a NOTIFY with an XML document each time the state changes. In IMS the UE and the P-CSCF both subscribe.

RFC 3680 · TS 24.229 §5.1.1.3

7.6 Keeping the registration alive



A registration is soft state. The device must refresh it before it expires, usually at about half the lifetime. In IMS the typical lifetime is 600 000 seconds, and the UE re-registers 600 seconds before expiry or when its IP address or access network changes.

RFC 3261 §10.2.4 · TS 24.229 §5.1.1.4

7.7 When registration fails

RESPONSE	LIKELY CAUSE
401 again and again	wrong password or key, clock or sequence problem
403 Forbidden	not subscribed, barred, roaming not allowed
404 Not Found	identity unknown in the HSS or registrar
408 / no answer	wrong proxy address, firewall, bearer missing
423 Interval Too Brief	Expires shorter than the registrar accepts
503 Service Unavailable	registrar or HSS overloaded, try another

A failed registration means no calls at all, so it is the first thing to check. The response code narrows it down quickly. 401 in a loop is a credential problem. 403 is a subscription or roaming problem. 404 says the user is unknown. Timeouts point at the network path or at the P-CSCF address.

RFC 3261 §21.4 · TS 24.229 §5.1.1

7.8 Three things to remember

- 1 REGISTER → 401 → REGISTER → 200**
The standard four-message registration.
- 2 The password never travels**
Only a hash over the nonce and the request.
- 3 No registration, no calls**
Check it first when a phone cannot be reached.

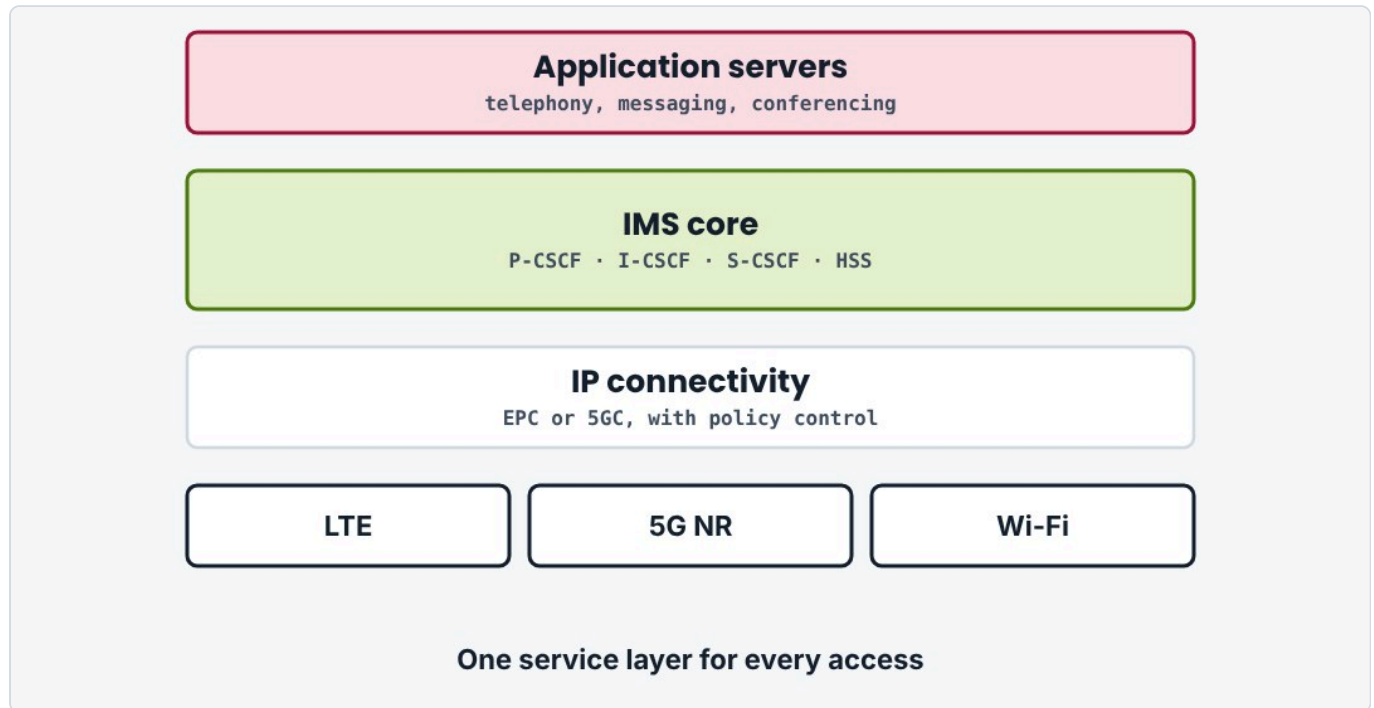
REGISTER binds the public name to the device. 401 is the normal first answer. The registration is soft state and must be refreshed.

RFC 3261 §10, §22

IMS Architecture

The IP Multimedia Subsystem: the three call session control functions, the HSS, application servers and the border to other networks.

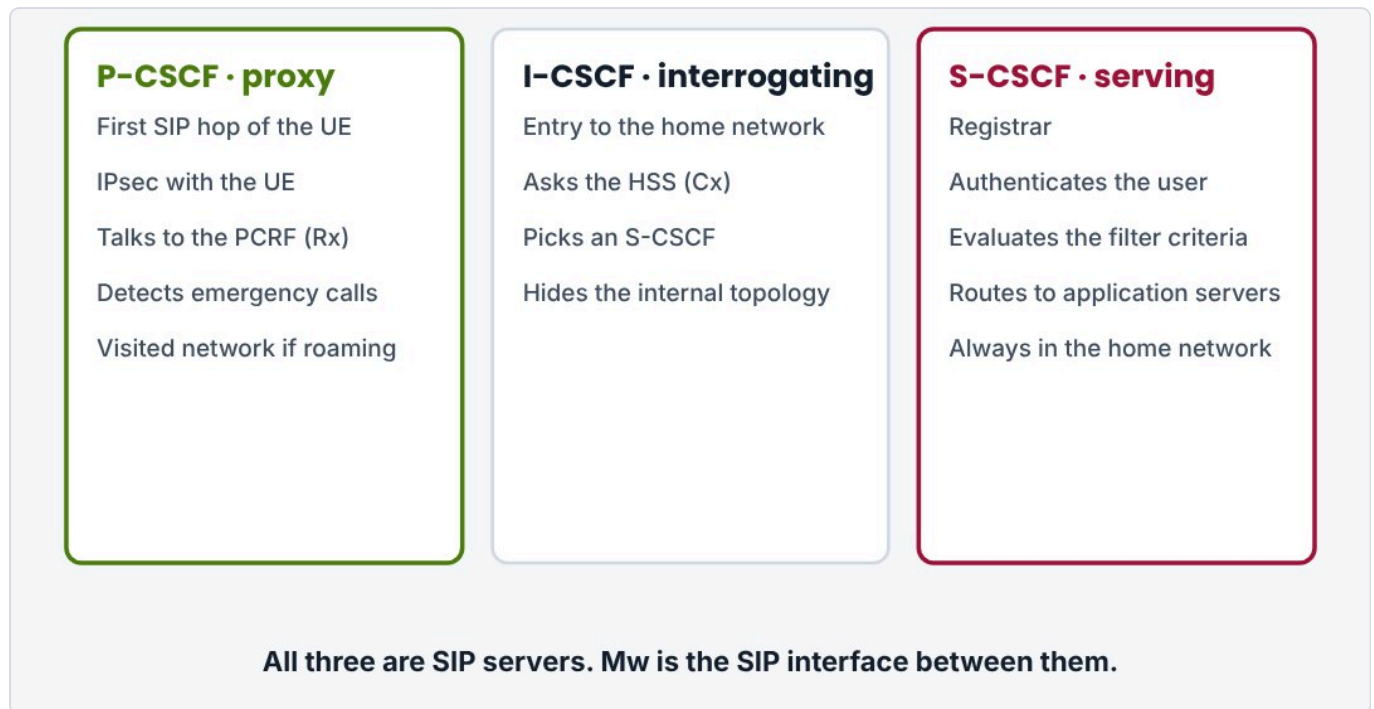
8.1 What IMS is



IMS is the 3GPP framework that delivers voice, video and messaging over IP. It is a set of SIP servers with defined roles, a subscriber database and application servers. It sits on top of any IP access: LTE, 5G or Wi-Fi. VoLTE, VoNR and VoWiFi are all IMS services.

TS 23.228 §4

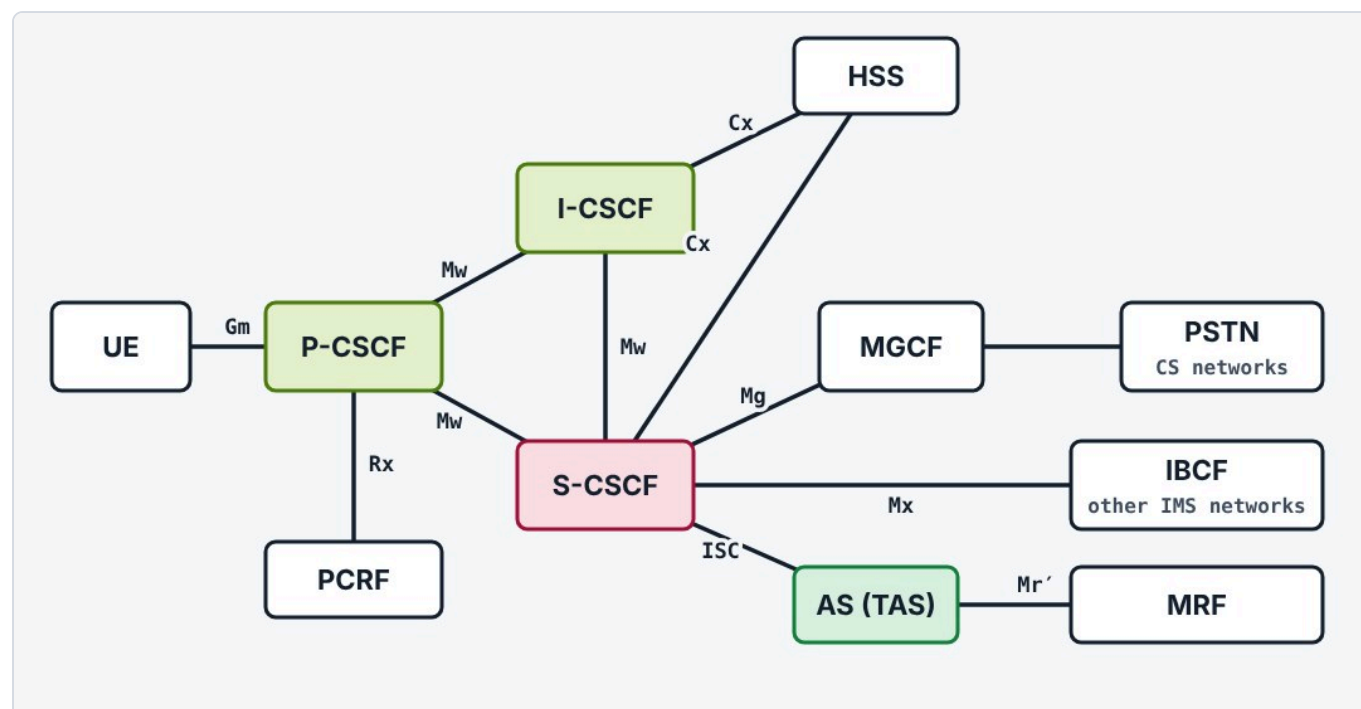
8.2 The three CSCFs



SIP signalling passes through three kinds of call session control functions. The proxy CSCF is the first contact point of the UE. The interrogating CSCF is the entry point of the home network and finds the right serving CSCF. The serving CSCF is the registrar and controls the session.

TS 23.228 §4.6

8.3 The complete picture



Around the CSCFs sit the supporting functions. The HSS holds the subscriber profile. Application servers provide the services. The MGCF and the media gateway connect to the circuit-switched telephone network. The IBCF is the border towards other IP networks.

TS 23.228 §4 · TS 23.002

8.4 Interfaces and protocols

INTERFACE	BETWEEN	PROTOCOL
Gm	UE – P-CSCF	SIP
Mw	CSCF – CSCF	SIP
ISC	S-CSCF – application server	SIP
Mg, Mj	S-CSCF, BGCF – MGCF	SIP
Cx	I-CSCF, S-CSCF – HSS	Diameter
Sh	application server – HSS	Diameter
Rx	P-CSCF – PCRF	Diameter
Ut	UE – application server	XCAP over HTTP

IMS has two families of interfaces. Everything that carries call signalling uses SIP. Everything that reads or writes subscriber data or policy uses Diameter. Knowing the letters lets you place a trace in the architecture at once.

TS 23.228 §4.4 · TS 29.228 · TS 29.214 · TS 24.229

8.5 Identities

IMPI · private identity

user@realm, used only for authentication

IMPU · public identity

sip:+4930123456@ims.example.com

or tel:+4930123456. A user can have several.

Implicit registration set

IMPUs that register together with one REGISTER

Derived from the IMSI

IMSI@ims.mnc.mcc.3gppnetwork.org

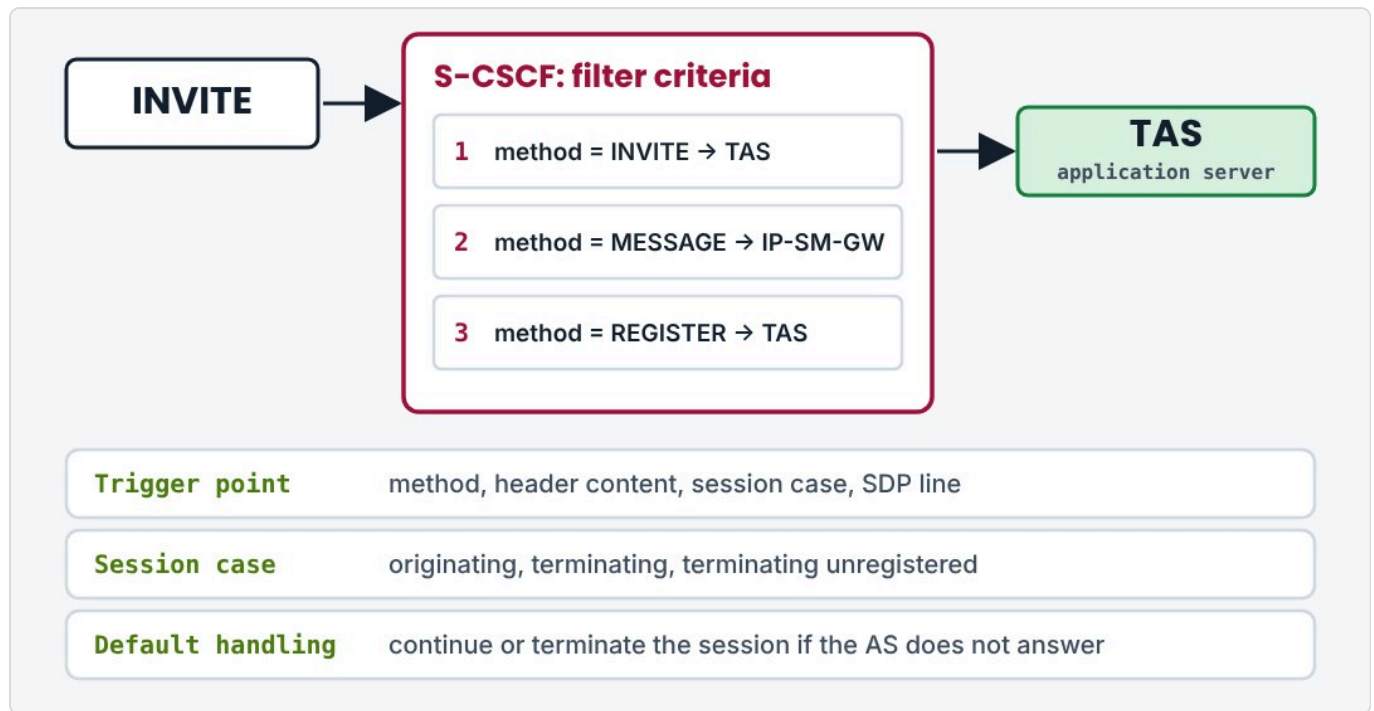
GRUU, IMEI

sip.instance with the IMEI identifies the device

An IMS user has one private identity and one or more public identities. The private identity is for authentication only and looks like a user name. The public identities are what other people dial: a SIP URI and usually a tel URI. They are stored on the ISIM, or derived from the IMSI when there is only a USIM.

TS 23.228 §4.3 · TS 23.003 §13

8.6 Service profile and filter criteria



The HSS stores a service profile for each user. Its core is a list of initial filter criteria. Each one says: if a request matches this trigger, send it to this application server. The S-CSCF downloads the list at registration and walks through it in priority order for every initial request.

TS 23.218 §5.2, §6 · TS 29.228 Annex B

8.7 The border elements

ELEMENT	JOB
P-CSCF + IMS-AGW	access border: NAT, IPsec, media anchoring, QoS
IBCF + TrGW	border to other IP networks: topology hiding, transcoding
BGCF	chooses where to break out to the CS network
MGCF + IM-MGW	SIP to ISUP/BICC, RTP to TDM
MRFC + MRFP	announcements, conferences, tones
E-CSCF + LRF	emergency calls and location

Networks do not expose their IMS core directly. Towards the UE, the P-CSCF is often combined with an access gateway into a session border controller. Towards other operators the IBCF and the TrGW do the same job. Towards the telephone network the MGCF translates SIP into ISUP and the media gateway converts the voice.

TS 23.228 §4.7, §4.14, Annex I · TS 29.163

8.8 Three things to remember

1

P, I and S

Proxy, interrogating and serving CSCF.

2

HSS over Cx

Authentication vectors, S-CSCF name, service profile.

3

Filter criteria

They decide which application server sees a request.

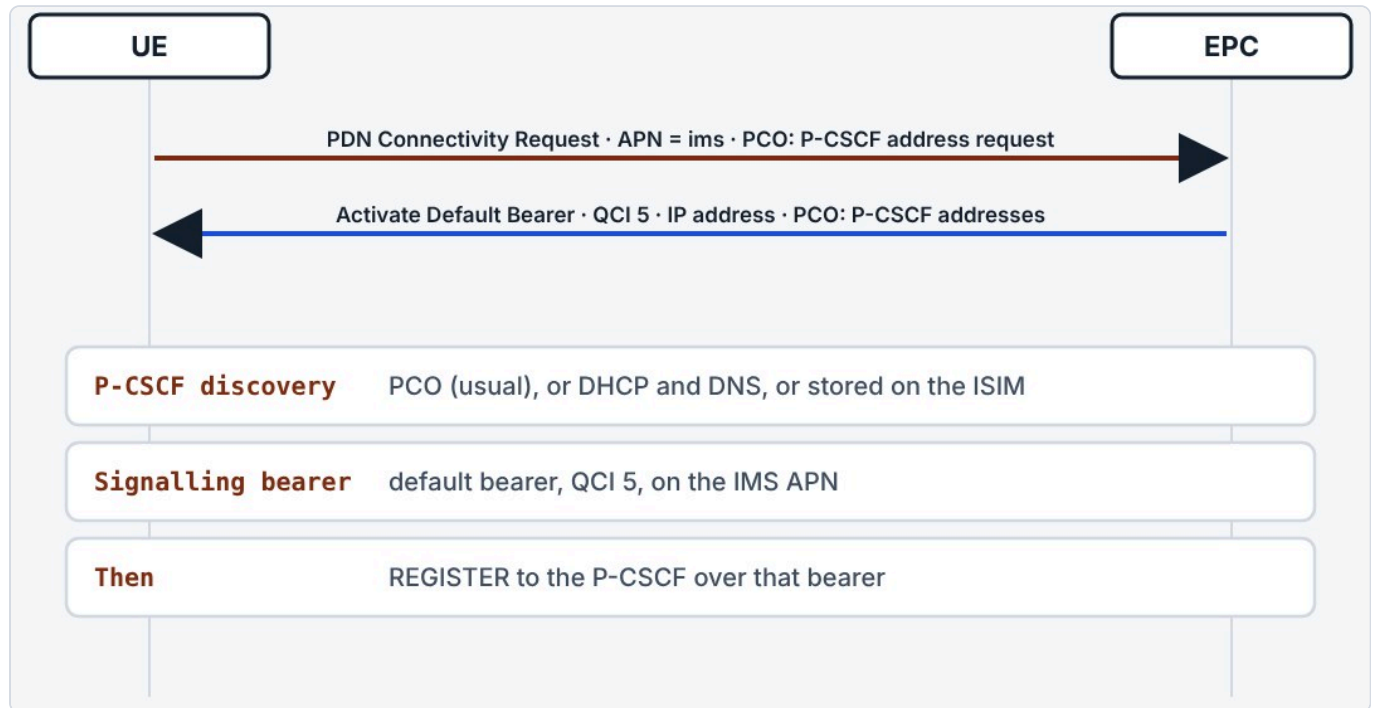
P-CSCF at the edge, I-CSCF at the door of the home network, S-CSCF in control. SIP for sessions, Diameter for data. Services live in application servers.

TS 23.228 §4

IMS Registration

From finding the P-CSCF to the second REGISTER: the complete IMS registration with AKA and IPsec.

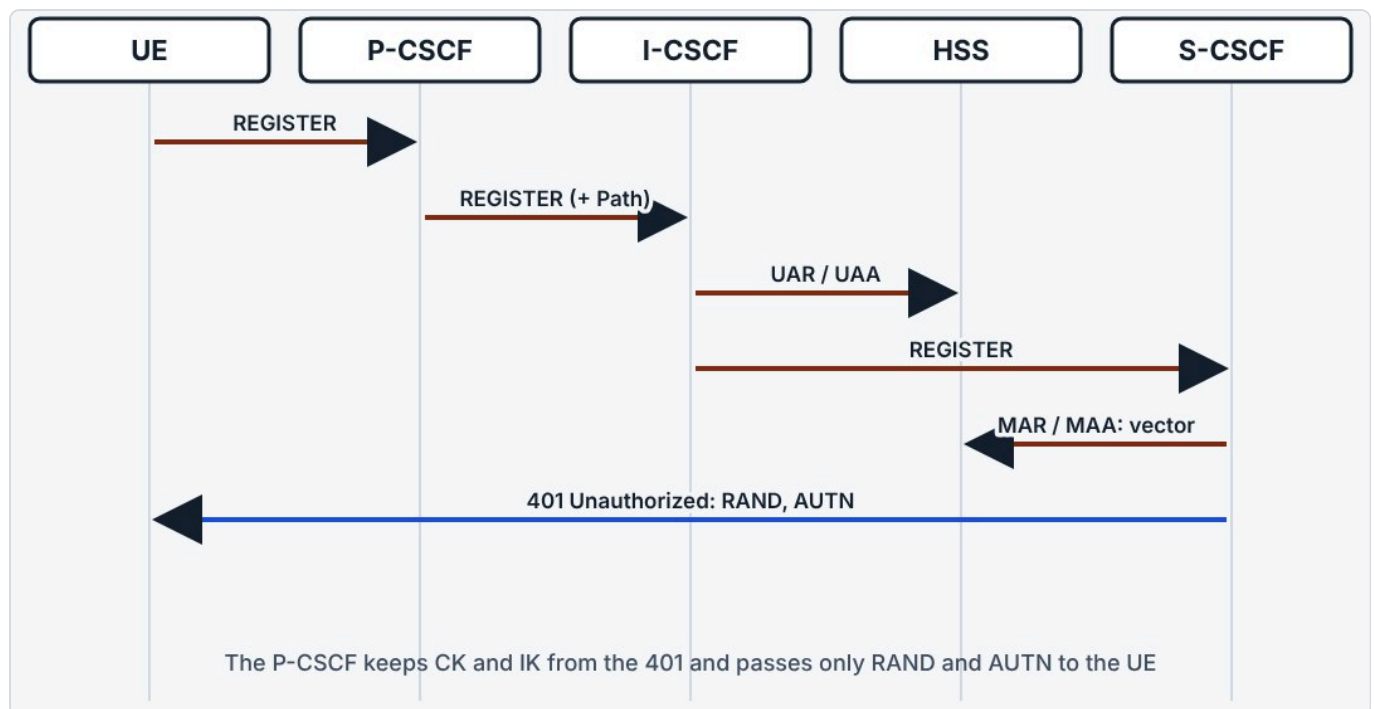
9.1 Before the first REGISTER



IMS registration needs IP connectivity first. The UE attaches, activates a default bearer on the IMS APN with QCI 5 and receives its IP address. In the same step it learns the address of the P-CSCF, usually from the protocol configuration options of the bearer activation.

TS 23.228 §5.1 · TS 24.229 §9.2.1, Annex L

9.2 The first REGISTER



The UE sends an unprotected REGISTER. The P-CSCF adds a Path header and forwards it to the I-CSCF of the home network. The I-CSCF asks the HSS which S-CSCF to use. The S-CSCF fetches an authentication vector and answers 401 with the challenge.

TS 23.228 §5.2.2.3 · TS 24.229 §5.1.1.2, §5.4.1.2 · TS 29.228

9.3 IMS AKA

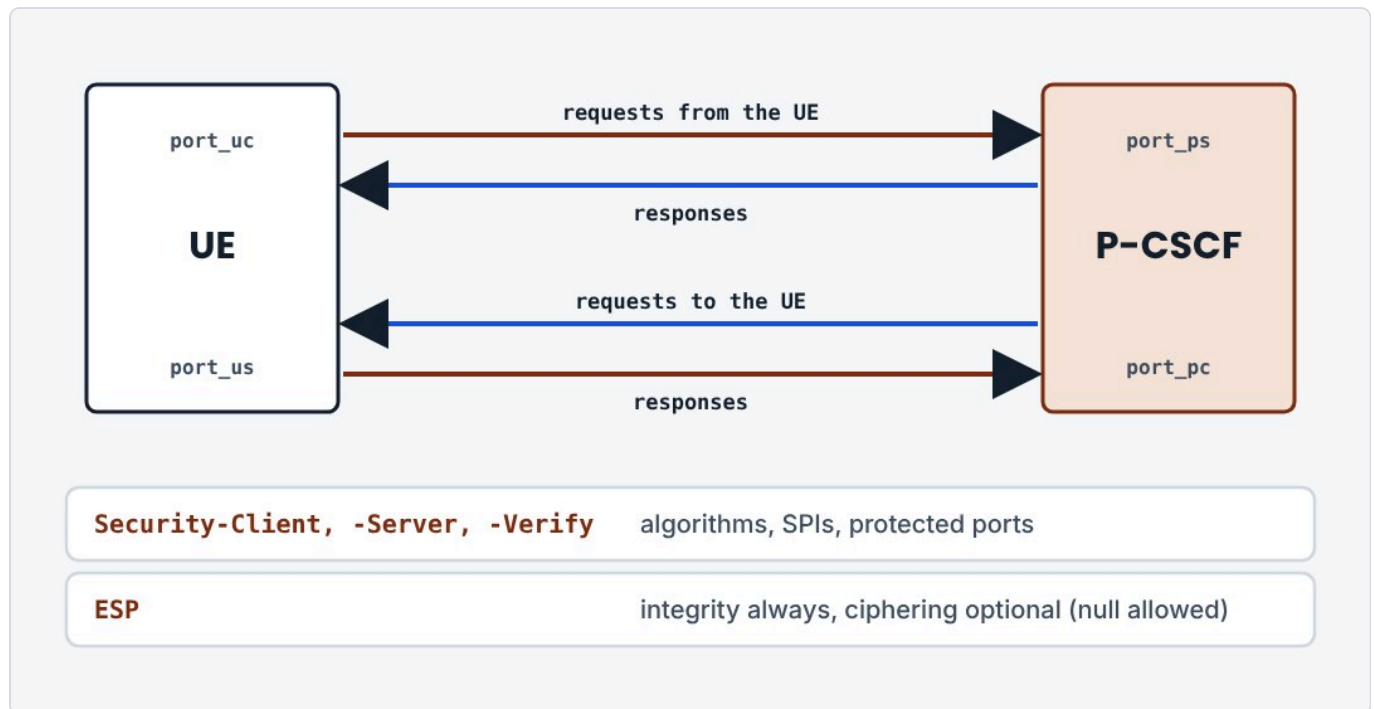
401 · WWW-Authenticate: Digest

Algorithm	AKAv1-MD5: digest framing, AKA content
nonce	carries RAND and AUTN, base64 coded
ISIM	verifies AUTN, computes RES, CK, IK
response	digest computed with RES as the password
CK, IK	become the keys of the IPsec tunnels
Sync failure	wrong sequence number: UE sends AUTS, S-CSCF fetches a new vector

IMS reuses the AKA method of the radio network. The ISIM checks AUTN, so the UE authenticates the network. It computes RES and the keys CK and IK. The RES goes into the second REGISTER as the digest response. The S-CSCF compares it with the expected value from the HSS.

TS 33.203 §6.1 · RFC 3310

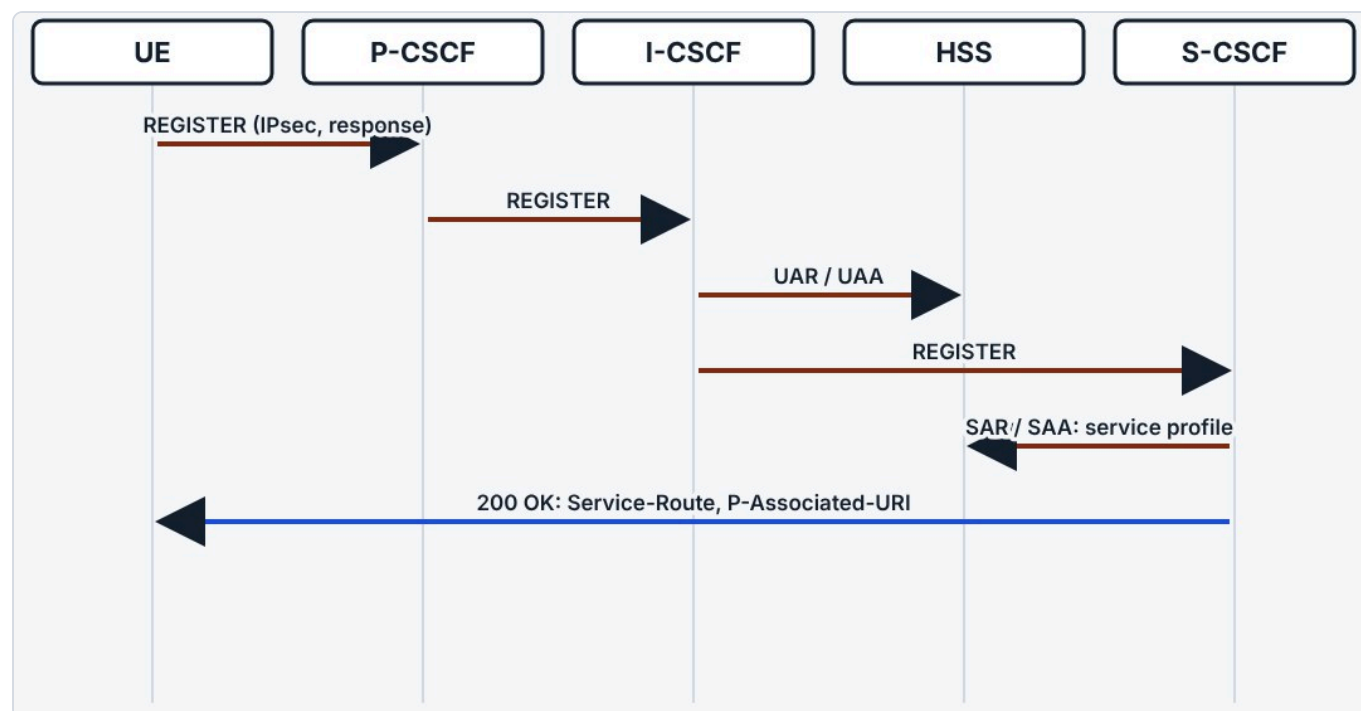
9.4 IPsec between UE and P-CSCF



The first REGISTER and the 401 also negotiate security. The UE offers algorithms and ports in Security-Client, the P-CSCF answers in Security-Server. With CK and IK both sides set up two pairs of IPsec ESP associations. From the second REGISTER on, all signalling runs inside them on the protected ports.

TS 33.203 §7 · RFC 3329 · TS 24.229 §5.1.1.5

9.5 The second REGISTER



The UE repeats the REGISTER, now protected and with the AKA response. The S-CSCF verifies it, tells the HSS that it now serves this user, and downloads the service profile. The 200 OK returns the Service-Route and the associated identities. The UE is registered.

TS 23.228 §5.2.2.3 · TS 24.229 §5.4.1.2.2

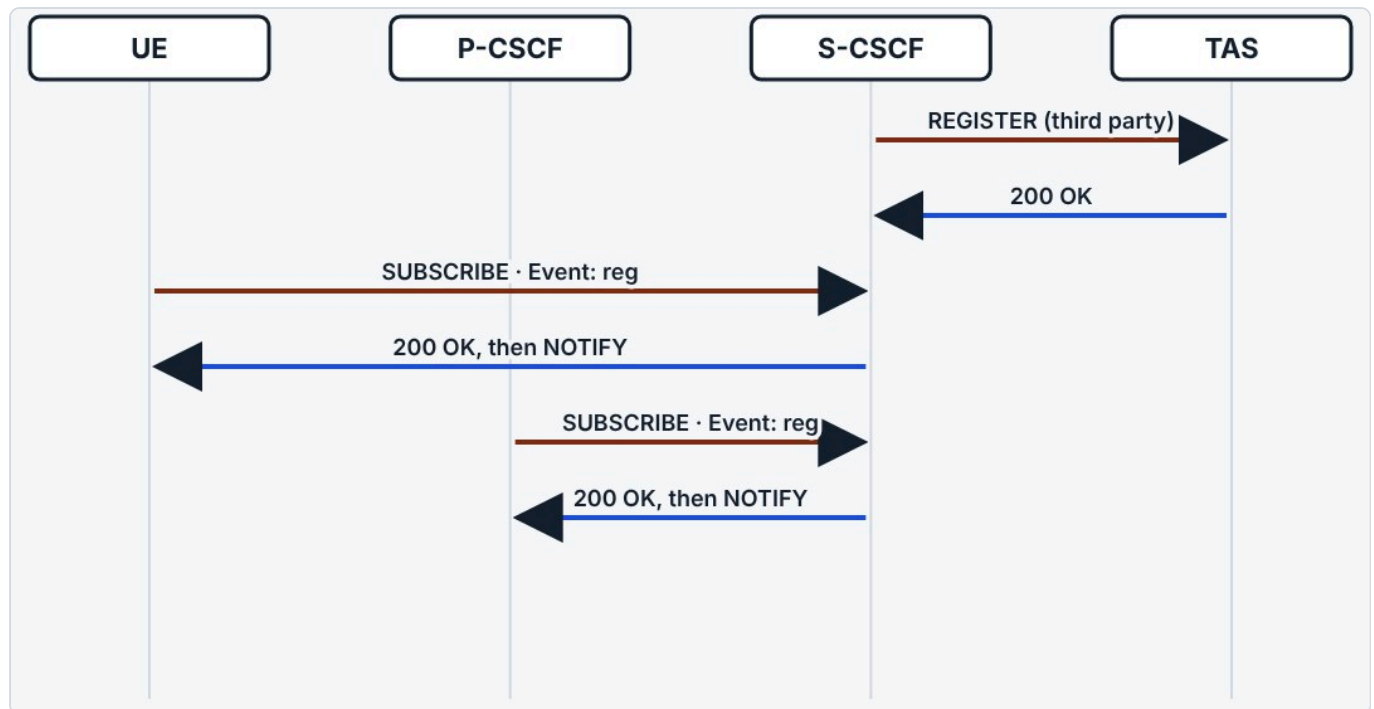
9.6 The Diameter messages on Cx

COMMAND	FROM	PURPOSE
UAR / UAA	I-CSCF	is the user allowed? which S-CSCF?
MAR / MAA	S-CSCF	get authentication vectors
SAR / SAA	S-CSCF	store the S-CSCF name, download the profile
LIR / LIA	I-CSCF	find the S-CSCF of a called user
RTR / RTA	HSS	network-initiated de-registration
PPR / PPA	HSS	push a changed profile

Three request and answer pairs on the Cx interface run the registration. They are worth knowing by name, because a failed registration often shows its real cause in the Diameter result code and not in the SIP response.

TS 29.228 §6.1 · TS 29.229

9.7 What follows registration



Registration triggers three more things. The S-CSCF performs a third-party registration towards the application servers named in the filter criteria. The UE and the P-CSCF subscribe to the registration state. And the phone shows the VoLTE symbol, because it can now make and receive calls.

TS 24.229 §5.4.1.7, §5.1.1.3, §5.2.3

9.8 Three things to remember

1 REGISTER → 401 → REGISTER → 200
Through P-CSCF, I-CSCF and S-CSCF.

2 AKA + IPsec
RES proves the UE, CK and IK protect the signalling.

3 UAR, MAR, SAR
Select the S-CSCF, get vectors, store and download.

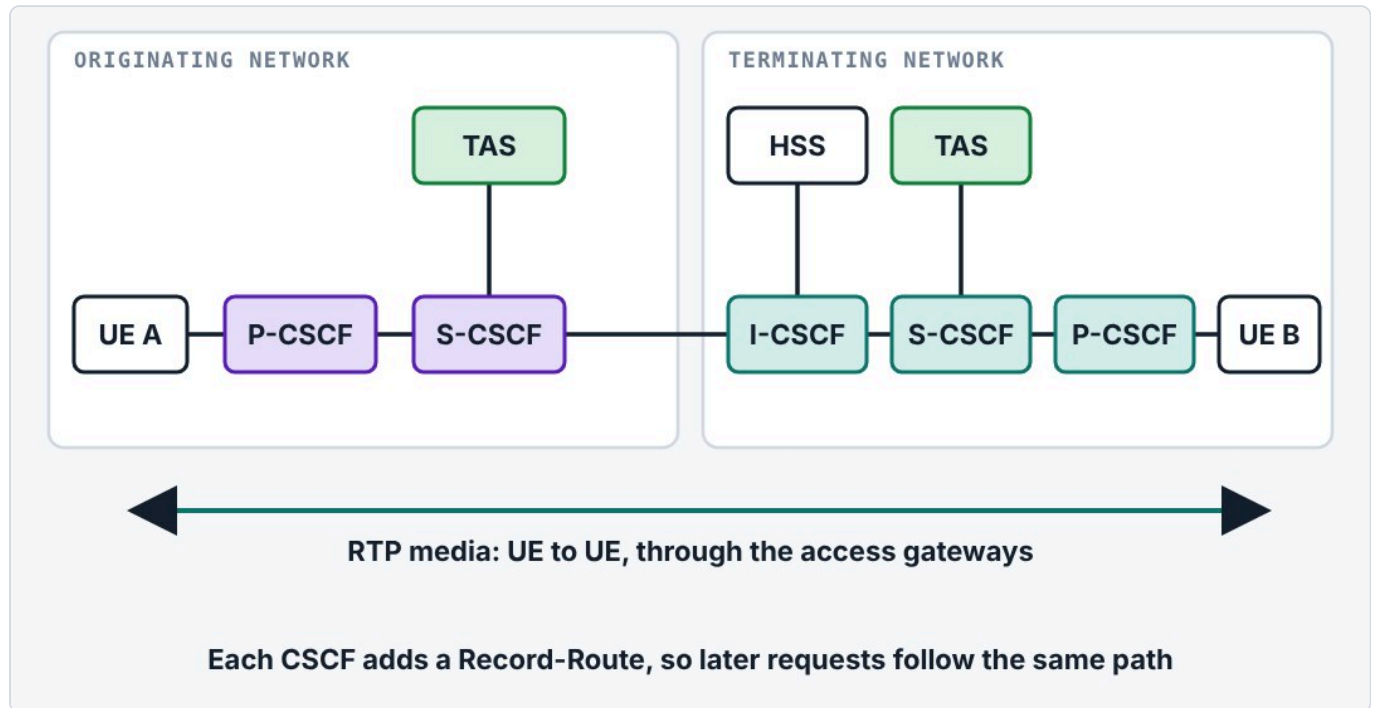
Two REGISTERs: the first is challenged, the second is protected. AKA gives authentication and the IPsec keys. Cx is where the HSS takes part.

TS 23.228 §5.2 · TS 33.203 · TS 24.229 §5.1.1

The VoLTE Call Flow

A complete IMS voice call: the signalling path, preconditions, the dedicated bearer and release.

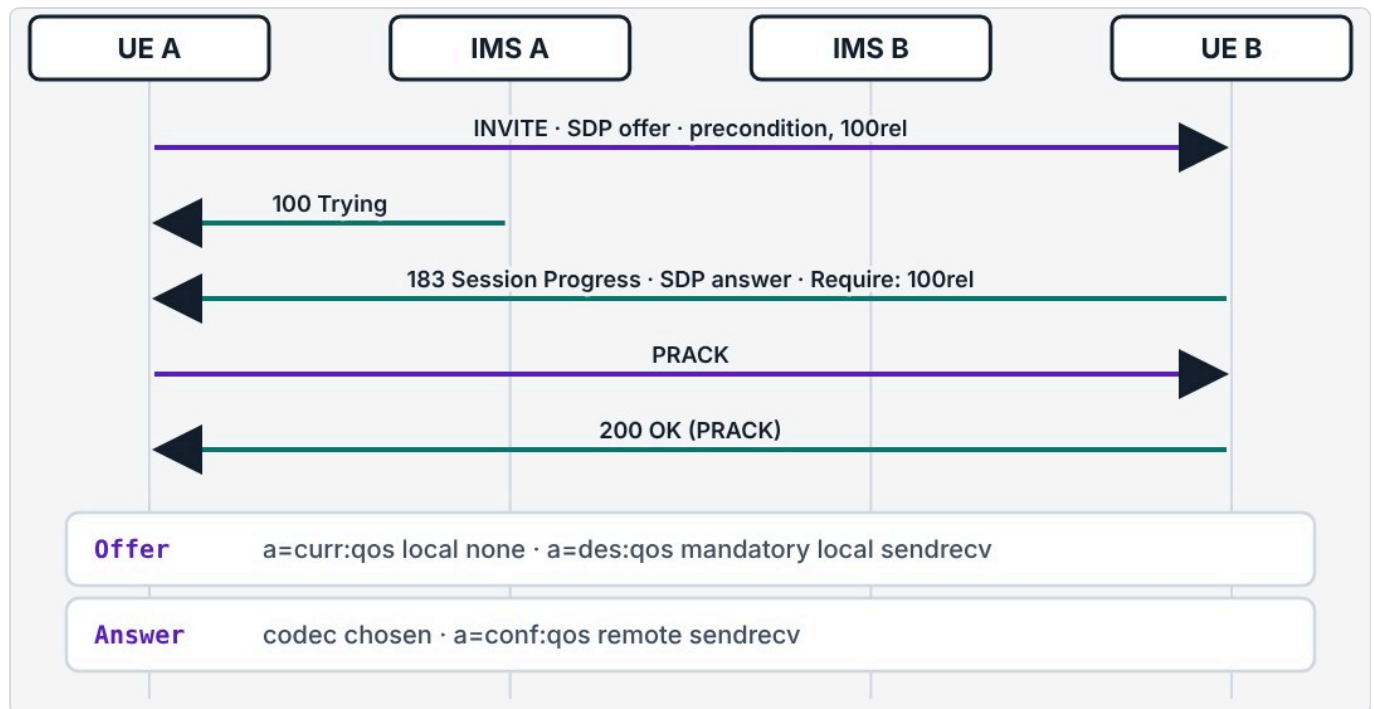
10.1 The signalling path



A call between two VoLTE users crosses both IMS networks. On the originating side the request passes the P-CSCF and the S-CSCF, which involves the telephony application server. The terminating network is entered through its I-CSCF, which finds the S-CSCF of the called user, and from there the request goes to the P-CSCF and the UE.

TS 23.228 §5.4, §5.5, §5.6

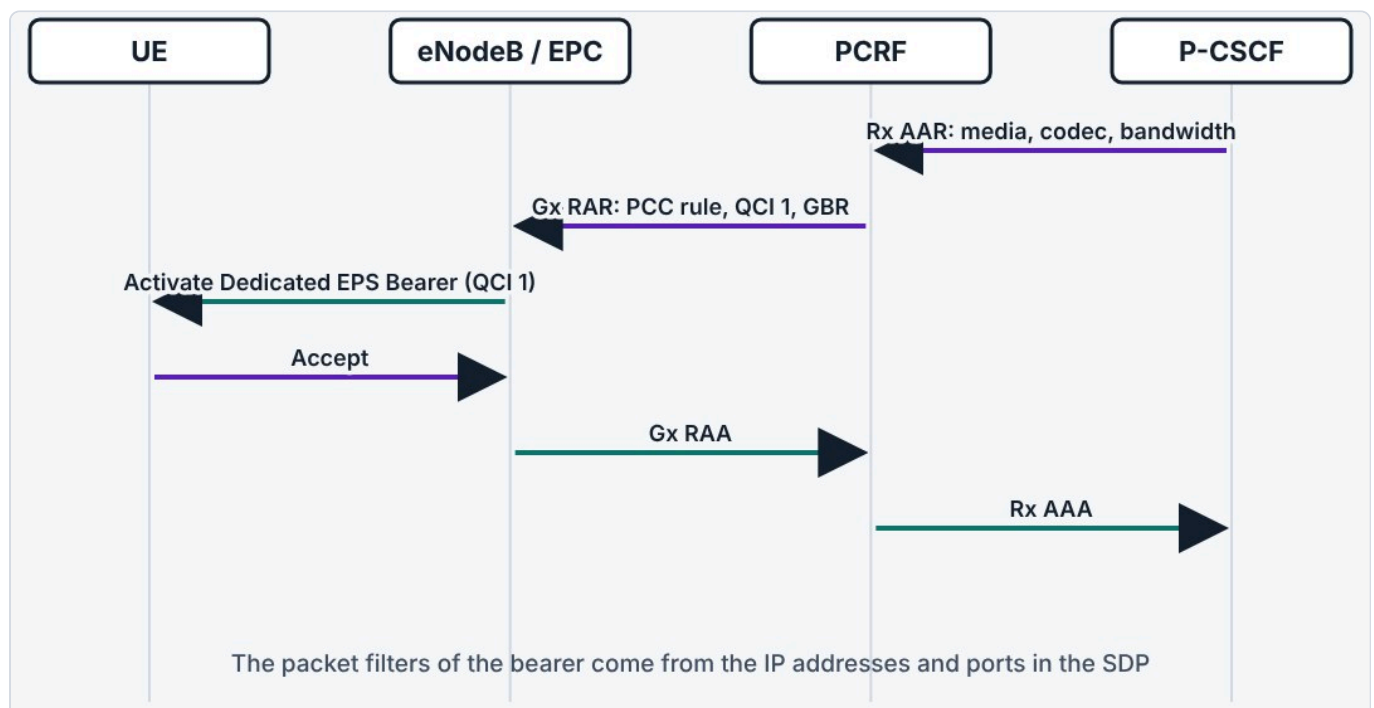
10.2 INVITE and 183: the SDP exchange



The INVITE carries the SDP offer with the codecs and the precondition lines, which say that no resources are reserved yet. The called UE answers with a reliable 183 Session Progress containing the SDP answer. It does not ring yet. The caller acknowledges the 183 with PRACK.

TS 24.229 §5.1.3, §6.1 · RFC 3262 · RFC 3312

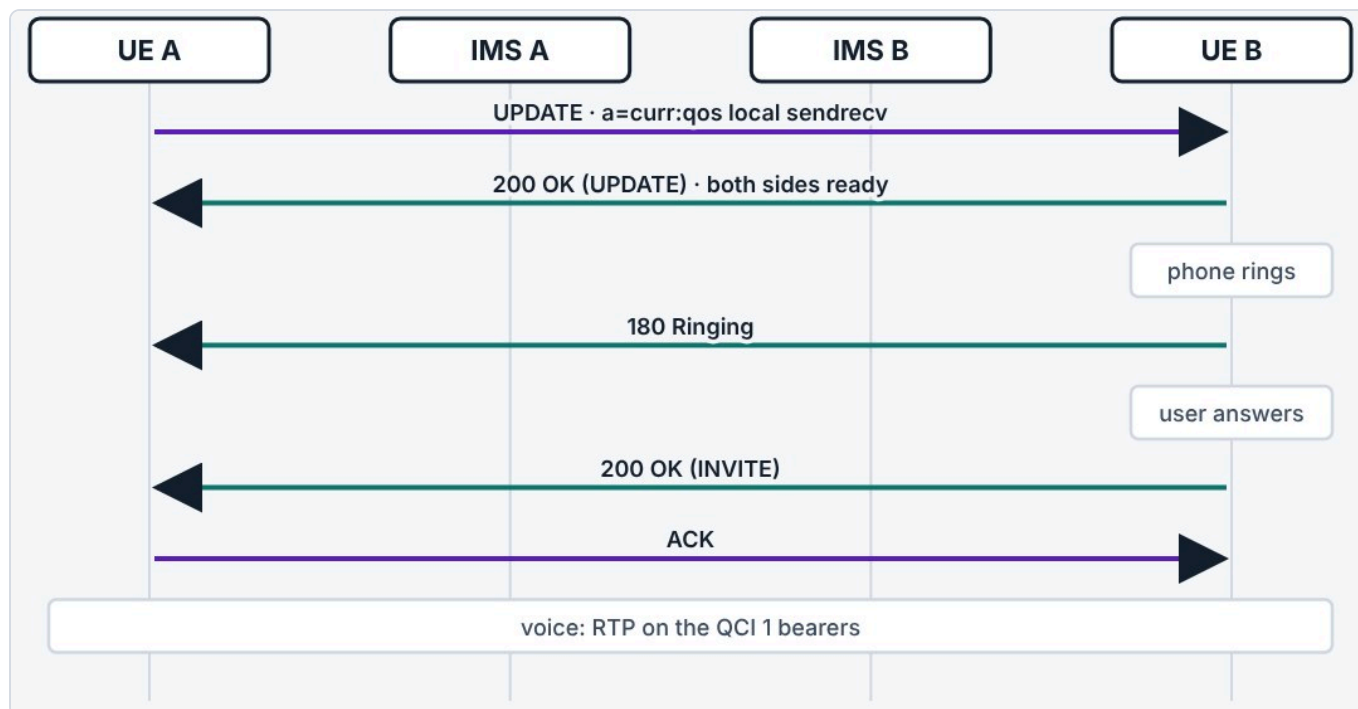
10.3 The network sets up the voice bearer



The SDP answer tells each P-CSCF which media will flow. It sends this to the PCRF over Rx. The PCRF turns it into a rule and the P-GW creates a dedicated bearer with QCI 1 and a guaranteed bit rate. This happens on both sides independently, each for its own UE.

TS 23.203 §6.1, §7.4 · TS 29.214 · TS 23.401 §5.4.1

10.4 UPDATE, ringing and answer



When its bearer is up, UE A sends UPDATE with the precondition now met on its side. UE B answers when its own bearer is ready too. Only now does UE B ring and send 180. When the user answers, the 200 OK and the ACK complete the call.

RFC 3311 · RFC 3312 §5 · TS 24.229 §5.1.3, §5.1.4

10.5 The whole flow on one page

#	MESSAGES	WHAT HAPPENS
1	INVITE → 100	SDP offer, preconditions not met
2	183 → PRACK → 200	SDP answer, codec agreed
3	(Rx, Gx, bearer setup)	QCI 1 bearer on both sides
4	UPDATE → 200	preconditions met
5	180 → (PRACK → 200)	alerting
6	200 → ACK	connected, media flows

Typical setup time VoLTE to VoLTE: about 1 to 3 seconds

Put together, a VoLTE call setup has five phases. Offer, answer, resource reservation, alerting and connect. It is longer than the basic SIP flow because the network guarantees that the voice bearer exists before the phone rings.

TS 24.229 · GSMA IR.92 · TS 24.237

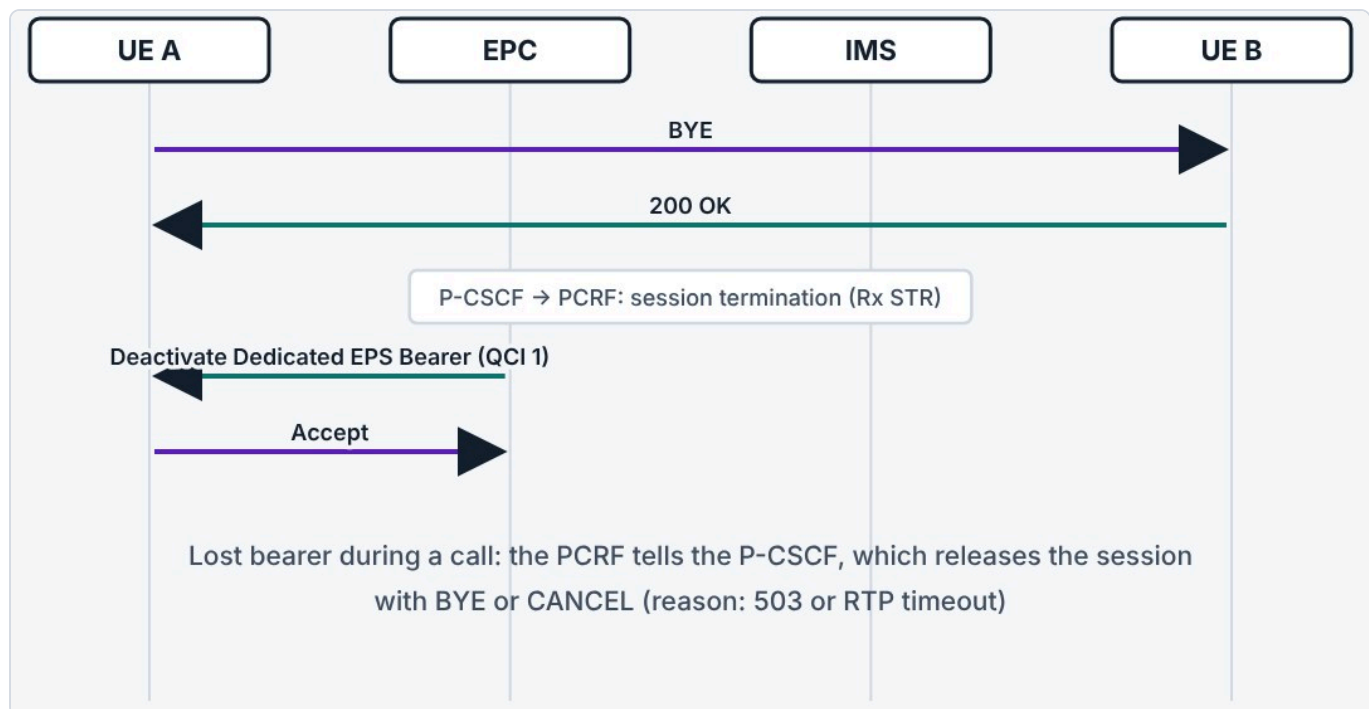
10.6 Headers to check in a VoLTE INVITE

P-Preferred-Identity	UE → P-CSCF: the identity I want to use
P-Asserted-Identity	after the P-CSCF: the verified caller identity
P-Access-Network-Info	3GPP-E-UTRAN-FDD; utran-cell-id-3gpp=...
Accept-Contact / Contact	+g.3gpp.icsi-ref="...mmtel": this is a telephony call
Supported / Require	100rel, precondition, timer
P-Early-Media	whether early media (ring-back) may be played

An IMS INVITE carries headers a plain SIP call does not have. They explain most interoperability problems. The identity headers say who is calling. The feature tags say that this is a telephony session. The Supported and Require headers show which extensions each side uses.

TS 24.229 §5.1.2A, §5.1.3 · GSMA IR.92 §2.2

10.7 Releasing the call



Either side ends the call with BYE. Each P-CSCF then tells the PCRF that the media is gone, and the network deletes the dedicated bearer. The default bearer with QCI 5 stays, and so does the registration. If the radio connection is lost instead, the network releases the session after the bearer loss is reported.

TS 23.228 §5.10 · TS 29.214 §4.4.4 · TS 23.401 §5.4.4

10.8 Three things to remember

1

INVITE, 183, PRACK, UPDATE, 180, 200, ACK

The VoLTE order of messages.

2

Rx and Gx link SIP to the bearer

P-CSCF → PCRF → P-GW → QCI 1.

3

No bearer, no ringing

Preconditions hold the call until QoS exists.

The SDP answer comes in the 183, not in the 200. The bearer is created by the network from the SDP. Ringing waits for both bearers.

TS 24.229 · TS 23.203 · RFC 3312

IMS Services

What runs on top of the IMS core: supplementary services, SMS, emergency calls, video and calls to other networks.

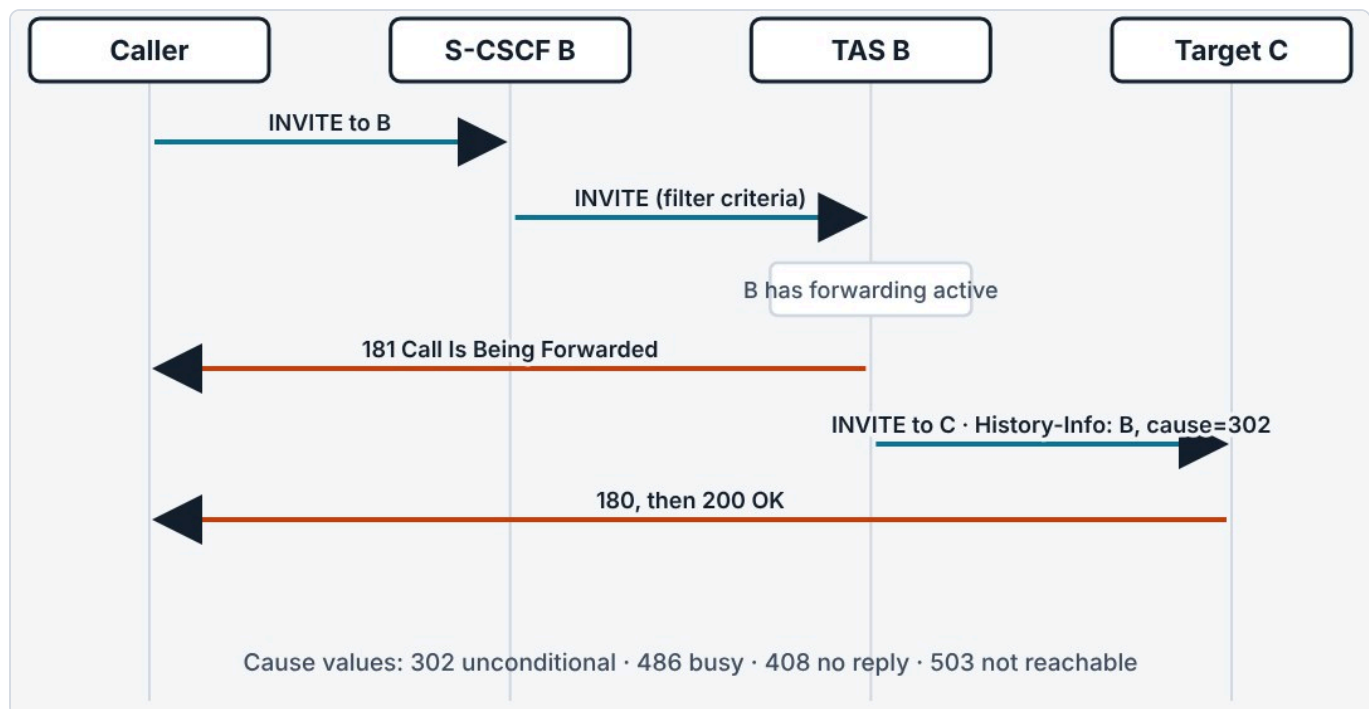
11.1 Supplementary services (MMTel)

SHORT	SERVICE	HOW IT SHOWS IN SIP
OIP / OIR	show or hide the caller identity	P-Asserted-Identity, Privacy: id
CDIV	call forwarding	181, History-Info, new INVITE
HOLD	call hold	re-INVITE, a=sendonly
CW	call waiting	180 with Alert-Info, or 182
CB	call barring	603 or 403 from the TAS
CONF	conference	INVITE to the conference URI, REFER

The telephony application server provides the services users know from the old telephone network. They are standardised as multimedia telephony. The user changes the settings, for example a forwarding number, over the Ut interface with XCAP, or with star codes sent as SIP.

TS 24.173 · TS 22.173 · TS 24.623

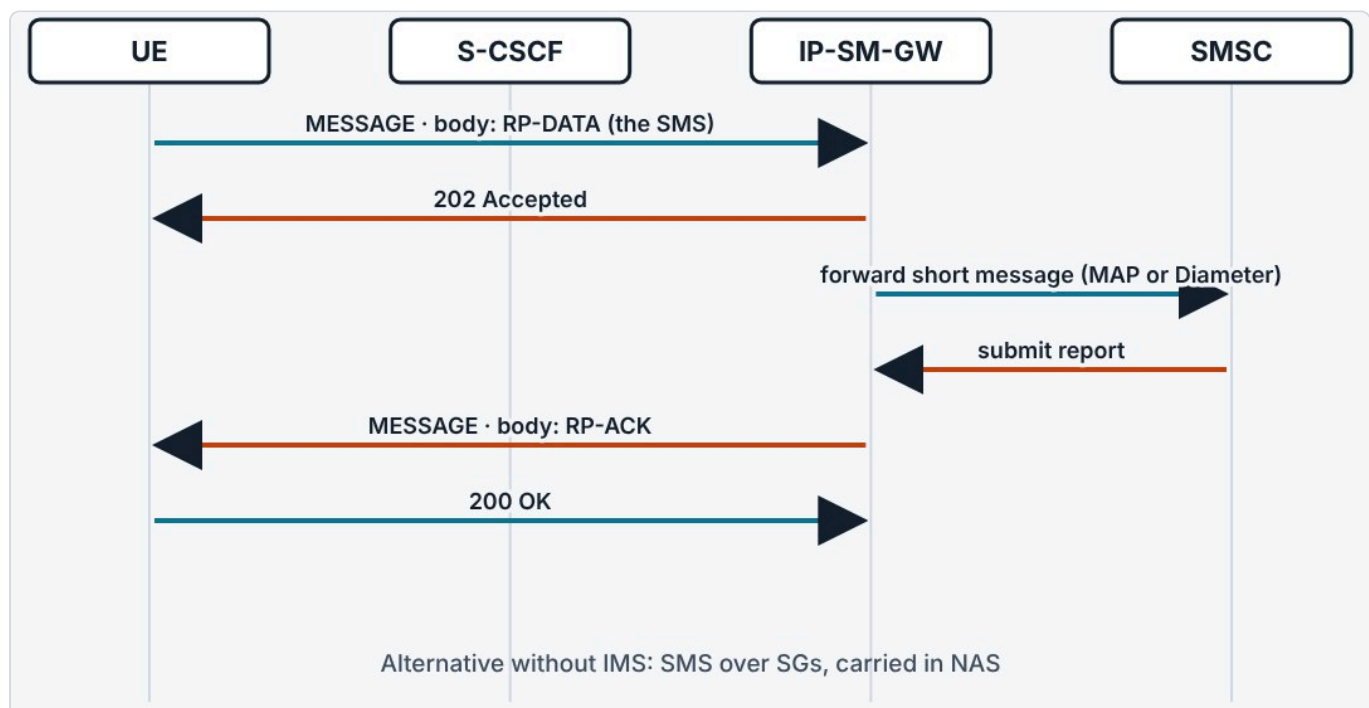
11.2 Call forwarding in a trace



Call forwarding is done by the TAS of the called user. It answers the caller with 181 Call Is Being Forwarded and sends a new INVITE to the forwarding target. The History-Info header records the original target and the reason, so the final recipient can see who was called first.

TS 24.604 · RFC 7044

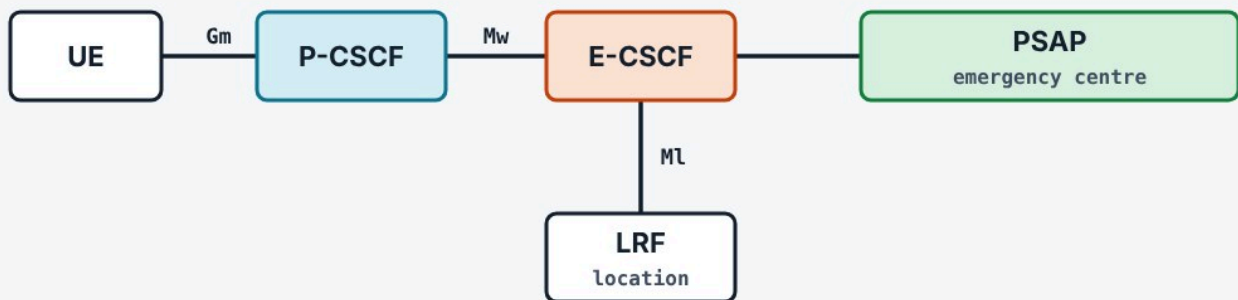
11.3 SMS over IP



An SMS can travel as a SIP MESSAGE. The body is the same binary short message as in 2G and 3G, with content type application/vnd.3gpp.sms. An IP short message gateway connects the IMS to the SMS centre. The delivery report comes back as a second MESSAGE.

TS 24.341 · TS 23.204 · RFC 3428

11.4 Emergency calls

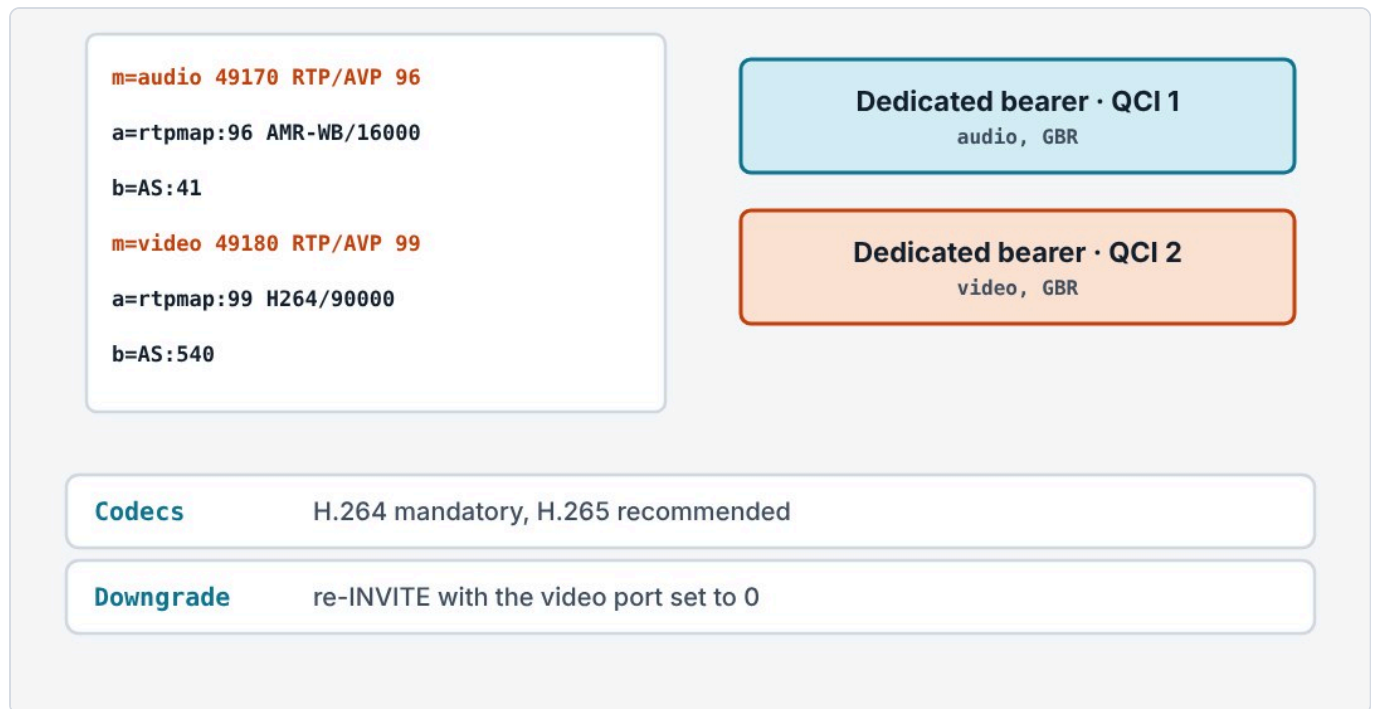


Request URI	urn:service:sos (or sos.police, sos.fire, sos.ambulance)
Bearer	emergency APN, QCI 5 signalling and QCI 1 voice, high ARP
No SIM	anonymous emergency call, no authentication

An emergency call must work without a normal registration and must reach the right emergency centre. The UE sets up an emergency bearer and performs an emergency registration. The P-CSCF recognises the emergency number and sends the INVITE to the E-CSCF, which uses the location to pick the public safety answering point.

TS 23.167 · TS 24.229 §5.1.6

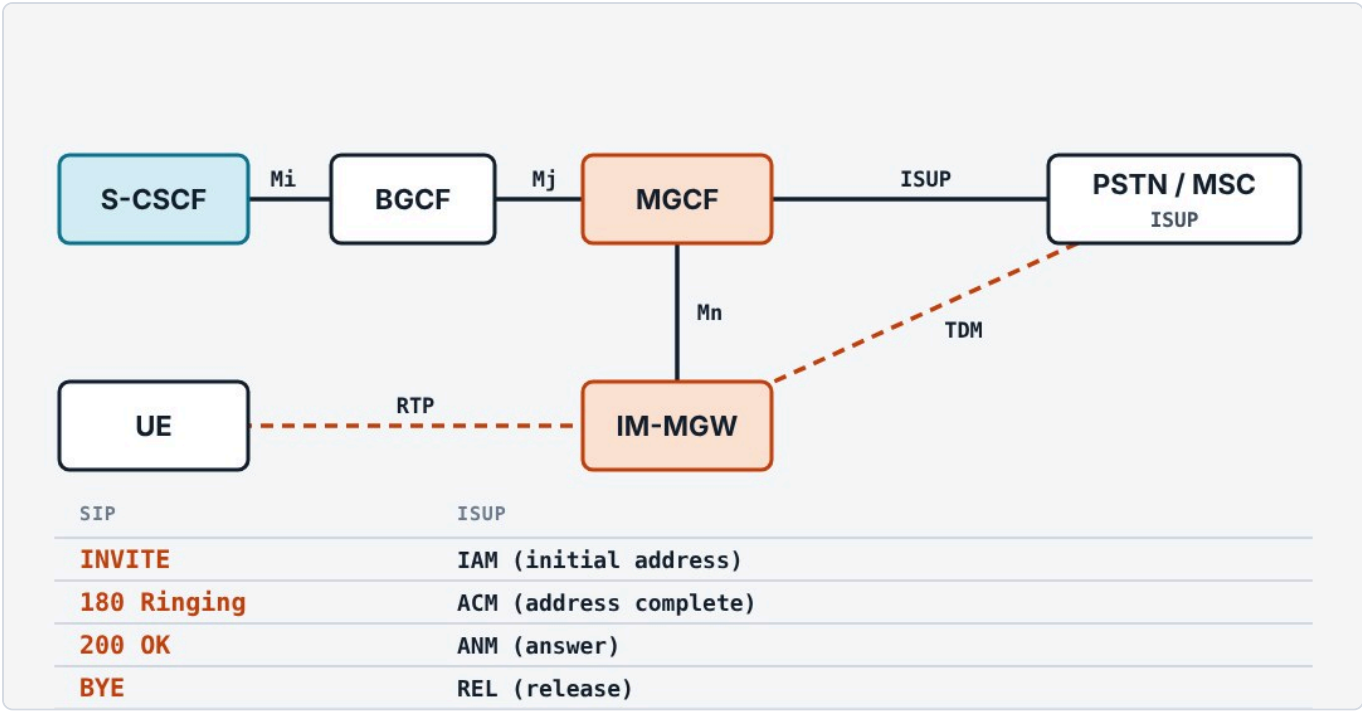
11.5 Video calls



A video call is a voice call with a second media line. The SDP offers audio and video, each with its own port and codec. The network creates two dedicated bearers: QCI 1 for the voice and QCI 2 for the video. Video can be added to or removed from a running call with a re-INVITE.

GSMA IR.94 · TS 26.114 §5.2.2

11.6 Calls to the telephone network



When the called number is not an IMS user, the S-CSCF hands the request to the BGCF. It chooses a media gateway control function, which translates SIP into the ISUP signalling of the circuit-switched network. The media gateway converts RTP into a 64 kbit/s channel and transcodes the voice if needed.

TS 23.228 §5.19 · TS 29.163

11.7 Roaming and interconnect

Local breakout (LBO)

P-CSCF and P-GW in the visited network
Visited network controls QoS
Needs IMS agreements between operators
Local emergency calls are simple

S8 home routing (S8HR)

P-GW and P-CSCF in the home network
Visited network only provides the bearer
Fast to roll out
Most common for VoLTE roaming today

Interconnect between IMS networks: IBCF ↔ IPX ↔ IBCF

Between operators, IMS traffic passes interconnection border functions on both sides, usually over an IP exchange network. For VoLTE roaming two models exist. With local breakout the P-CSCF is in the visited network. With S8 home routing everything, including the P-CSCF, stays in the home network.

TS 23.228 §4.2.3, §4.15 · GSMA IR.65, IR.88

11.8 Three things to remember

1

TAS = supplementary services

Forwarding, barring, hold, conference.

2

SMS is a SIP MESSAGE

With the classic short message inside.

3

urn:service:sos

Emergency calls take their own path through the E-CSCF.

Services live in the application servers. The same SIP methods carry calls, messages and service control. Border functions connect IMS to everything else.

TS 24.173 · TS 24.341 · TS 23.167

Transport, Security, NAT and Troubleshooting

What carries SIP, how it is protected, why NAT causes trouble, and a method for reading a trace.

12.1 Transport: UDP, TCP, TLS

TRANSPORT	PORT	NOTES
UDP	5060	no connection, SIP timers do the retransmission
TCP	5060	reliable, needed for large messages
TLS	5061	encrypted hop by hop, sips: scheme
SCTP	5060	rare, between network nodes
WebSocket	80 / 443	SIP from a browser (WebRTC gateways)

Rule: larger than 1300 bytes, or within 200 bytes of the MTU → use TCP

SIP runs over UDP or TCP on port 5060, and over TLS on port 5061. UDP is simple, but SIP must then retransmit by itself, and large messages get fragmented. A message close to the path MTU must therefore be sent over TCP. IMS messages are long, so VoLTE phones switch between UDP and TCP all the time.

RFC 3261 §18 · TS 24.229 §4.2A

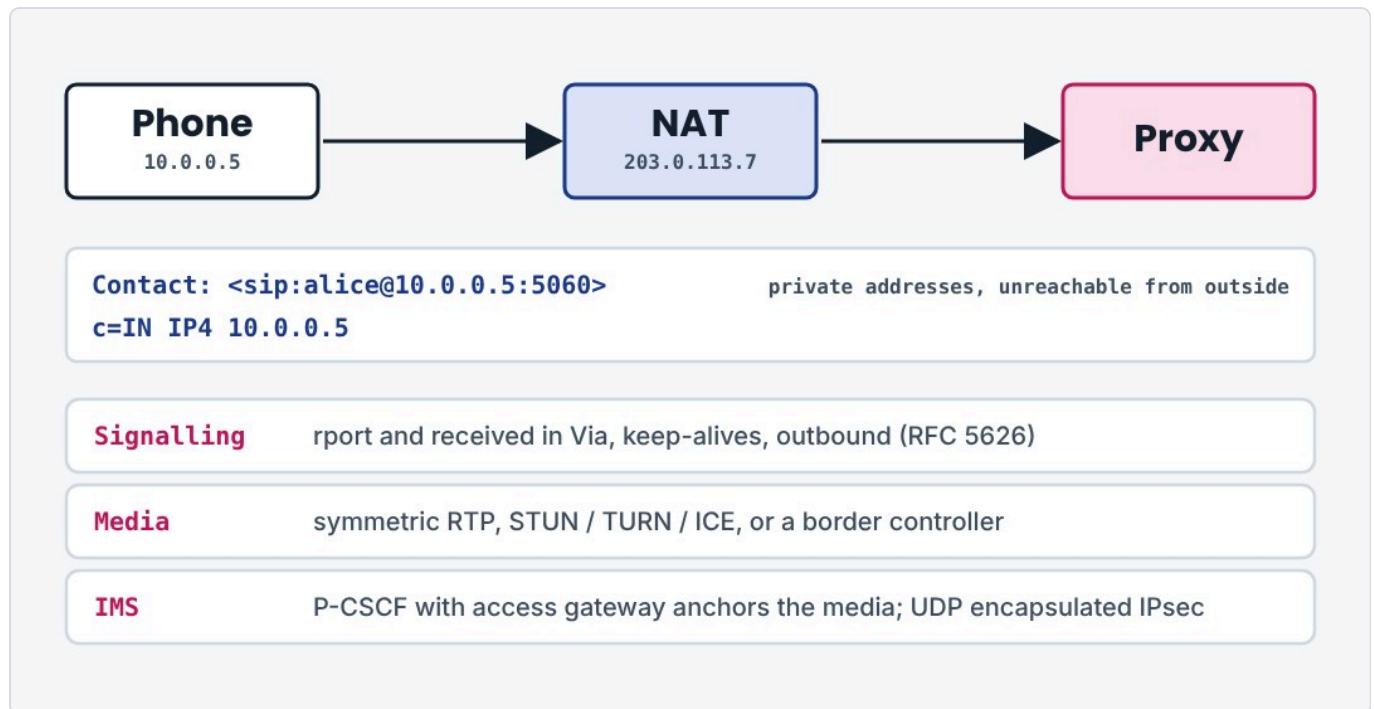
12.2 Security in layers

Authentication	Digest (password) or IMS AKA (ISIM)
Signalling	hop by hop: TLS, or IPsec ESP between UE and P-CSCF
Media	SRTP, keys exchanged with SDES or DTLS-SRTP
Asserted identity	P-Asserted-Identity inside the trust domain
Privacy	Privacy: id removes the identity at the border
Caller ID validation	STIR/SHAKEN: a signed Identity header

SIP security is built from several pieces. Digest or AKA authenticates the user. TLS or IPsec protects the signalling hop by hop. SRTP protects the media. Inside an operator network the nodes trust each other, and the P-Asserted-Identity header carries the verified identity across that trust domain.

RFC 3261 §26 · TS 33.203 · RFC 3711 · RFC 3325

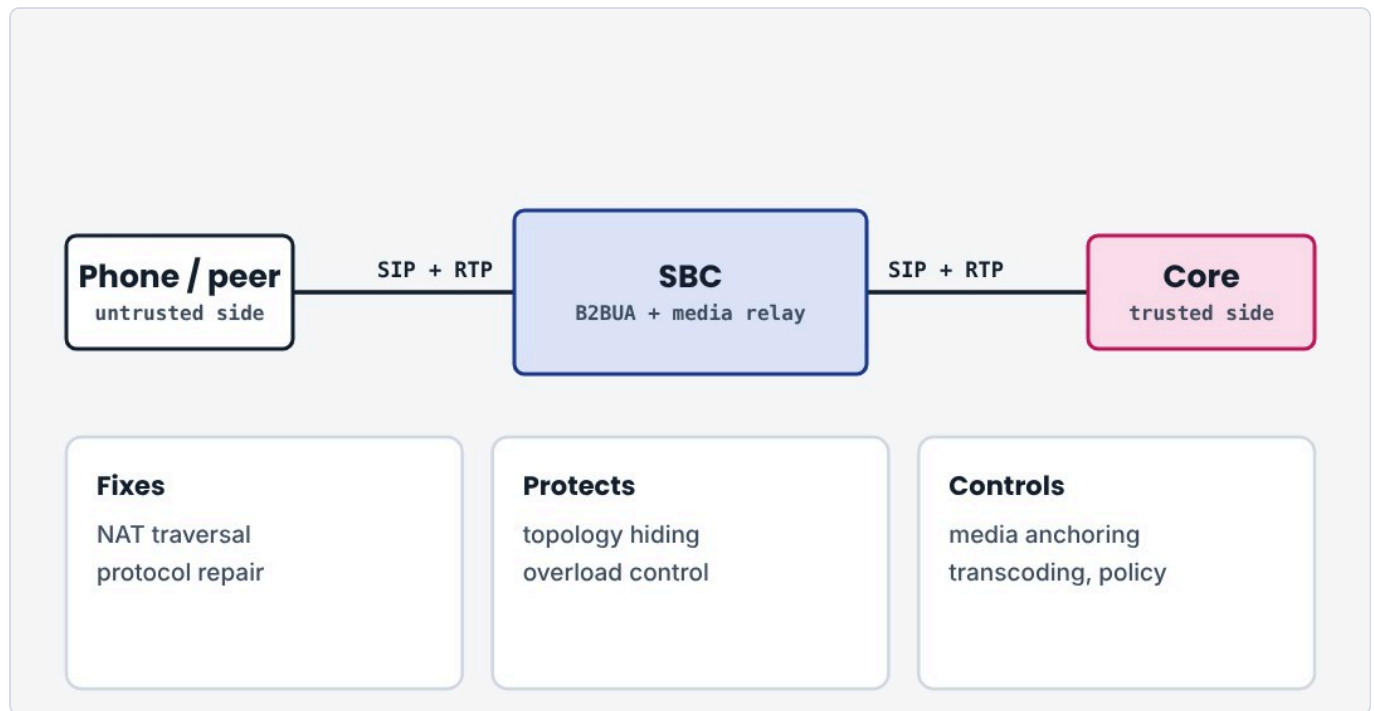
12.3 Why NAT breaks SIP



A device behind a NAT writes its private address into Via, Contact and the SDP. The other side cannot reach that address. Signalling answers and incoming calls get lost, and the media flows one way or not at all. The NAT also forgets the mapping after a short idle time.

RFC 3581 · RFC 5626 · RFC 8445

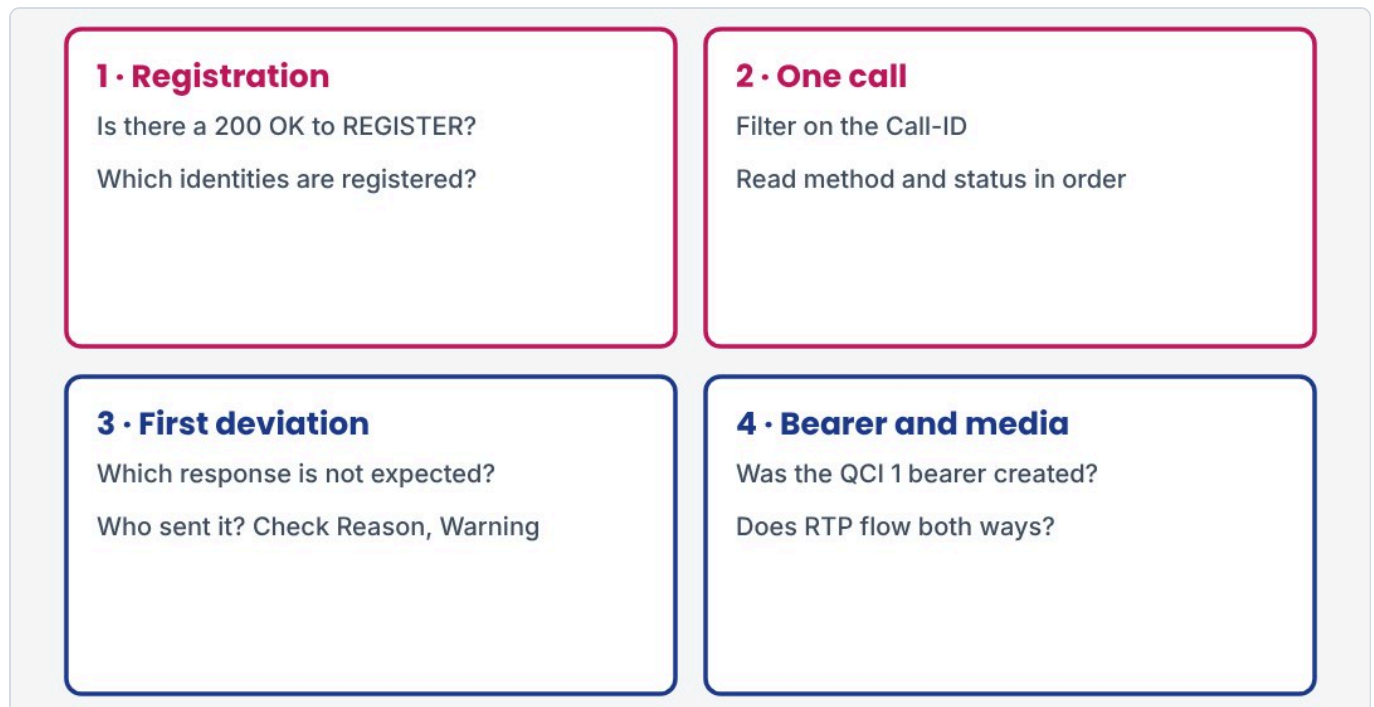
12.4 Session border controllers



In practice the border controller solves most of these problems. It is a back-to-back user agent at the edge of the network. It ends the signalling and the media on one side and starts them again on the other. That lets it fix addresses, hide the topology, enforce policy and transcode.

RFC 5853 · TS 23.228 Annex G

12.5 Reading a trace: where to start



A method saves time. Start with the registration, because nothing works without it. Then filter on one Call-ID and read the start lines in order. Find the first message that is not the expected one. Then check the bearer and the media before you go deeper into the headers.

RFC 3261 • TS 24.229

12.6 Failure patterns

WHAT YOU SEE	LOOK AT
403 to INVITE	registration lost, barring, identity not allowed
404, 604	wrong number format, ENUM or routing
408, no answer at all	next hop unreachable, timer B or F expired
480	called user not registered or not reachable
488, 606	SDP: codec, IP version, preconditions
500, 503	node failure, overload, or the QCI 1 bearer failed
CANCEL after 183	caller gave up, or bearer setup took too long
BYE with Reason	RTP timeout, radio link lost, SRVCC

Most call failures fall into a few patterns. The response code together with the node that sent it tells you where to look next. The Reason and Warning headers often carry the precise cause, including the Q.850 cause from the circuit-switched side.

RFC 3326 • RFC 3261 §20.43 • TS 24.229

12.7 Timers worth knowing

TIMER	TYPICAL VALUE	WHAT IT GUARDS
T1 (IMS, towards UE)	2 s	first retransmission
Timer B / F (IMS)	128 s	INVITE and non-INVITE transaction
Session-Expires	1800 s	a hung session
Registration	600 000 s	the binding in the S-CSCF
No-answer (TAS)	about 20 to 30 s	forwarding on no reply
RTP inactivity	10 to 30 s	lost media, releases the call

Many releases in a trace are caused by timers. If you know their values you can tell an expected timeout from a fault. The SIP transaction timers are longer in IMS. On top come the session timer, the registration lifetime and, in the UE, the timer that waits for the voice bearer.

RFC 3261 Table 4 · RFC 4028 · TS 24.229 Table 7.7.1

12.8 Three things to remember

1

Authentication + hop security + SRTP

Three separate layers of protection.

2

The SBC sits at every border

It is a B2BUA, so Call-ID and tags may change there.

3

Registration, Call-ID, first deviation

A fixed order makes trace reading fast.

UDP for small messages, TCP for large ones. NAT and borders rewrite addresses. In a trace, find the first unexpected message.

RFC 3261 §18, §26